Malagasy runs of homozygosity

2024-11-07

```
res_dir <- "../results/s07_ROH/"
```

Load gt

```
malagasy <-
  gt_load("../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree.gt")
```

This script is for the analysis of runs of homozygosity (ROH) in the Malagasy population, using the Malagasy only dataset containing 243 individuals and 3 loci.

Examine SNP density in our dataset

```
density <- show_loci(malagasy) %>%
  group_by(chromosome, window = floor(position / 1000)) %>% # per Kb
  summarise(n = n(), .groups = "drop")

min_snp_coverage <- min(density$n)
cat("Minimum SNP coverage: ", min_snp_coverage, "\n")
```

```
## Minimum SNP coverage:  1
```

```
median_snp_coverage <- median(density$n)
cat("Median SNP coverage: ", median_snp_coverage, "\n")
```

```
## Median SNP coverage:  1
```

```
mean_snp_coverage <- mean(density$n)
cat("Mean SNP coverage: ", mean_snp_coverage, "\n")
```

```
## Mean SNP coverage:  1.179967
```

Set up parameters

```
window_size <- 50 # size of window in SNPs, average SNP coverage is 1 per Kb, so this is approx 50Kb
threshold <- 0.05 #
min_snp <- 50 # min n. SNPs per run, average SNP coverage is 1 per Kb, so this is approx 50Kb
heterozygosity <- FALSE
max_opp_window <- 1 # n. of heterozygous SNPs permitted per run
max_miss_window <- 5 # max number of missing permitted
max_gap <- 100000 # max distance between SNPs in BP to be a run
min_length_bps <- 500000 # minimum length in BP of a run, 500kb
min_density <- 0.02 # minimum n. of SNP per kbps, roughly 1 SNP per 50kb
```

`windows_indiv_roh` uses a sliding window method to identify runs of homozygosity. Window size was set to 50 SNPs (approximately 50 considering median SNP density in the Malagasy dataset). This is the upper window limit tested by (Pemberton et al. 2012). Minimum length of runs was set to 5×10^5 BPs, with a maximum distance of 10^5 between successive SNPs in a window and a minimum SNP density of 0.02 per kilobase. Within each window, the thresholds for number of missing SNPs was set to 5 and the maximum number of heterozygous SNPs permitted was set to 1. These parameters were chosen to match the methods of (Kirin et al. 2010).

```
malagasy <- malagasy %>% group_by(population)
runs <- windows_indiv_roh(malagasy,
  window_size = window_size,
  threshold = threshold,
  min_snp = min_snp,
  heterozygosity = heterozygosity,
  max_opp_window = max_opp_window,
  max_miss_window = max_miss_window,
  max_gap = max_gap,
  min_length_bps = min_length_bps,
  min_density = min_density
)
```

```
# Load fastmixture Q matrix
qmats <- readRDS("../data/intermediate/fastmixture/fastmixture_m.RData")
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.gt")
# subset to only malagasy
all_malagasy <- c(
  "Betsimisaraka", "Merina", "Sakalava_Tsimihety",
  "Betsileo", "Diego", "Vezo", "Mikea", "Temoro"
)
# tidy with the gt
tidyq <- augment(get_q_matrix(qmats, k = 3, run = 1), data = mad_bor_g1000_sum)
malagasyq <- tidyq %>% filter(population %in% all_malagasy)
# subset the malagasyq dataframe to only the "Merina"
merina <- malagasyq %>% filter(population == "Merina")
# group the merina
merina$group <- ifelse(merina$.Q2 < 0.5, "Social group A", "Social group B")
merinaA <- merina %>% filter(group == "Social group A")
merinaB <- merina %>% filter(group == "Social group B")

# match the runs$Individual and malagasy$id
matches <- match(runs$id, malagasy$id)
# add $population to runs
runs$Population <- malagasy$population[matches]
runs$Population <- as.character(malagasy$population[matches])
# if $Individual is in merinaA$id, add "merina_a" to runs$Population
runs$Population <- ifelse(runs$id %in% merinaA$id, "Merina_A", runs$Population)
# if $Individual is in merinaB$id, add "merina_b" to runs$Population
runs$Population <- ifelse(runs$id %in% merinaB$id, "Merina_B", runs$Population)
head(runs)
```

```
##           group       id chrom nSNP      from        to lengthBps    Population
## 1 Betsimisaraka BET001RR     1  129  72419029  73592003   1172974 Betsimisaraka
## 2 Betsimisaraka BET001RR     1  127  82665976  83336086    670110 Betsimisaraka
## 3 Betsimisaraka BET001RR    10  476  19426984  20892774   1465790 Betsimisaraka
```

```
## 4 Betsimisaraka BET001RR    11  276  46282817  48732982   2450165 Betsimisaraka
## 5 Betsimisaraka BET001RR    11  151 107113002 107715515    602513 Betsimisaraka
## 6 Betsimisaraka BET001RR    12  302  18961806  20064514   1102708 Betsimisaraka
```

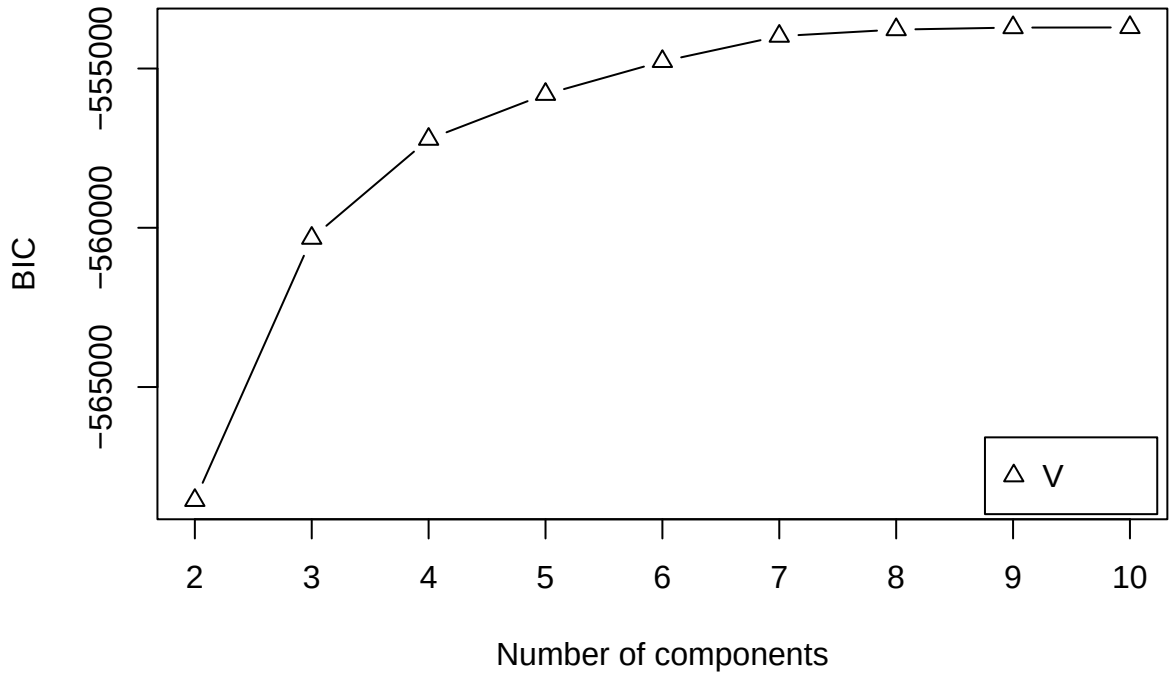
Classify the runs

A Gaussian mixture-model from the package mclust (Scrucca et al. 2023) was then used to classify runs into categories by length. Similarly to the methods of (Pemberton et al. 2012), a range of 2 to 10 clusters were investigated, and BIC was used to find the optimum number of clusters through the clustCombiOptim function.

See https://rdrr.io/cran/mclust/man/clustCombi.html for details.

This method is similar to GARLIC (Szpiech, Blant, and Pemberton 2017), used by (Korunes et al. 2022), which uses a Gaussian mixture model to classify runs into 3 categories of small, medium, and long. Importantly, when GARLIC clusters runs across populations, the Gaussian mixed-model alters the thresholds classification of runs - so that in one population the threshold for a 'short' run will differ from the threshold in another population. However, with mclust the thresholds are fixed across populations, so that a 'short' run in one population is equivalent to a 'short' run in another population.

```
G <- 2:10
BIC <- mclust::mclustBIC(data = runs$lengthBps, G = G, modelNames = "V")
plot(BIC) # shows BIC values for G 1:10, plateaux at 3 clusters
```




```
summary(BIC)
```

```
## Best BIC values:
##              V,10          V,9          V,8
```

```
## BIC      -553710.6 -5.537145e+05 -553777.12504
## BIC diff       0.0 -3.945119e+00     -66.55295
```

```
classes_optimum <- mclust::Mclust(
  data = runs$lengthBps,
  G = 1:10, modelNames = "V"
)

clust_classes <- clustCombi(object = classes_optimum, data = runs$lengthBps) #

optimum <- clustCombiOptim(clust_classes) # returns optimum number of clusters
optimum$numClusters.combi
```

```
## [1] 4
```

```
classes <- mclust::Mclust(
  data = runs$lengthBps,
  G = optimum$numClusters.combi
) # creates the model
runs$classification <- as.factor(classes$classification)
classes$BIC # BIC values for the model, when given G
```

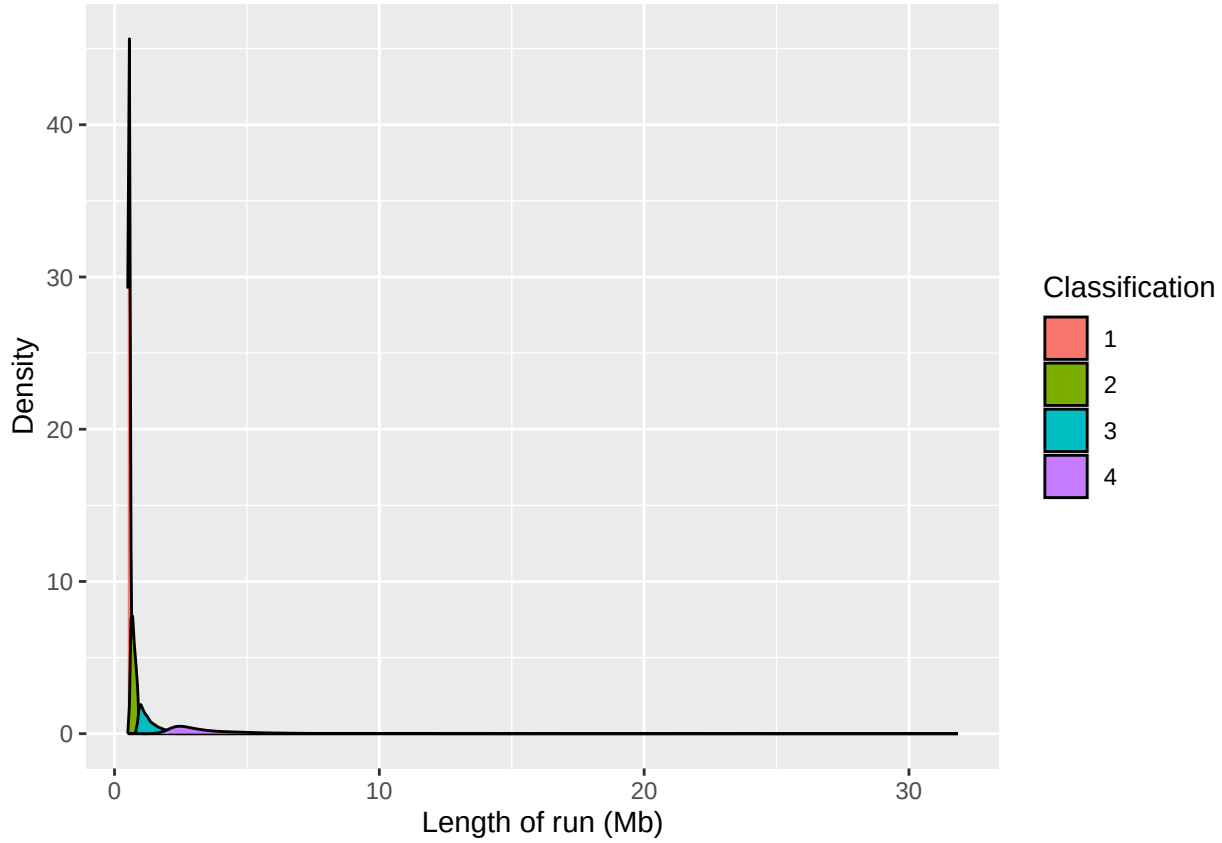
```
## Bayesian Information Criterion (BIC):
##           E         V
## 4 -593794.4 -557219.1
##
## Top 3 models based on the BIC criterion:
##       V,4       E,4
## -557219.1 -593794.4
```

```
classes$parameters # gives details of each class
```

```
## $pro
## [1] 0.24675990 0.40158618 0.28109042 0.07056349
##
## $mean
##         1         2         3         4
##  545702.8  700624.3 1161543.9 3519527.0
##
## $variance
## $variance$modelName
## [1] "V"
##
## $variance$d
## [1] 1
##
## $variance$G
## [1] 4
##
## $variance$sigmasq
## [1] 9.312104e+08 1.070379e+10 1.199834e+11 7.251766e+12
##
## $variance$scale
## [1] 9.312104e+08 1.070379e+10 1.199834e+11 7.251766e+12
```

```
ggplot(runs, aes(x = lengthBps / 1000000, fill = classification)) +
  geom_density() +
```

```
  labs(x = "Length of run (Mb)", y = "Density", fill = "Classification")
```

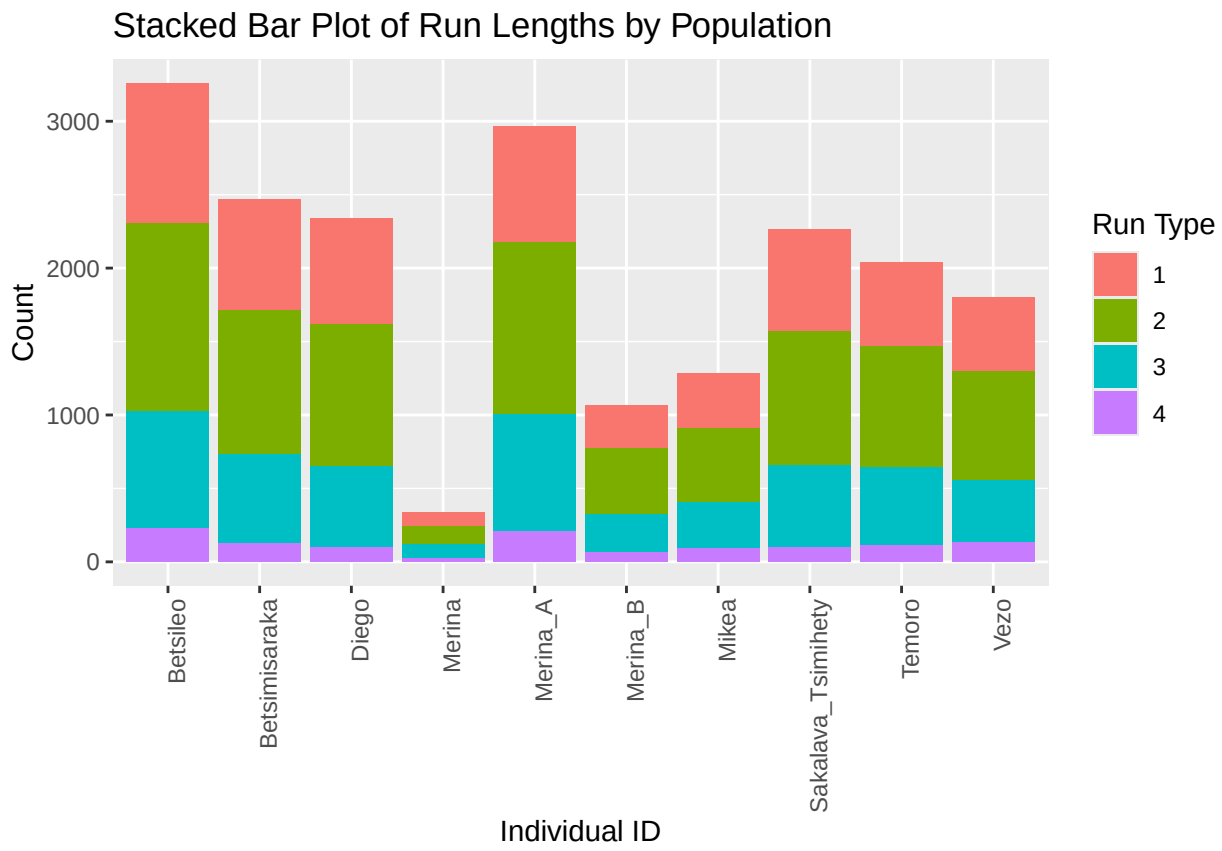


Plot runs by class

```
runs_per_indiv <- runs %>%
  group_by(id, classification) %>%
  count()

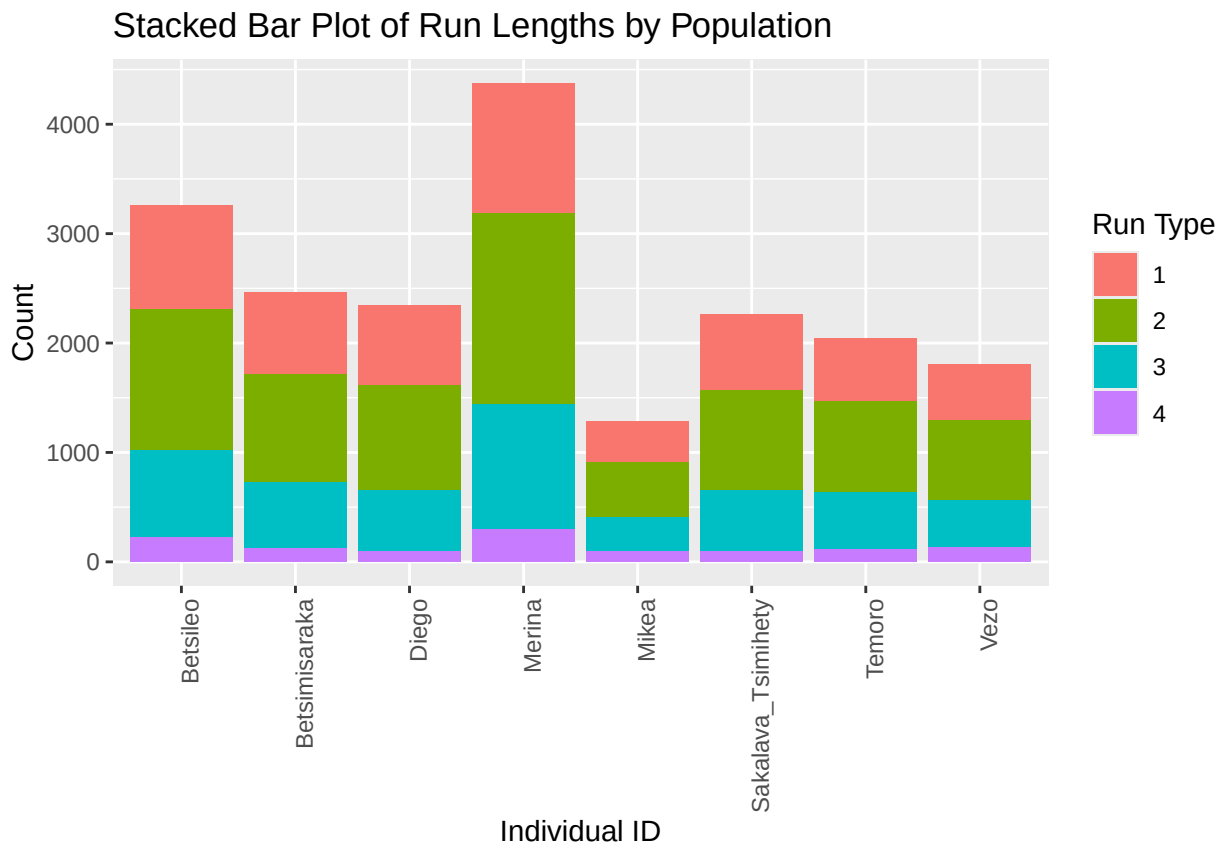
# add runs$population to runs_per_indiv
runs_per_indiv$population <- runs$Population[match(runs_per_indiv$id, runs$id)]


ggplot(runs_per_indiv, aes(x = population, y = n, fill = classification)) +
  geom_bar(stat = "identity", position = "stack") + # Stack bars instead of dodging
  labs(
    title = "Stacked Bar Plot of Run Lengths by Population",
    x = "Individual ID",
    y = "Count",
    fill = "Run Type"
  ) +
  # 90 degree hjust on axis text
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```

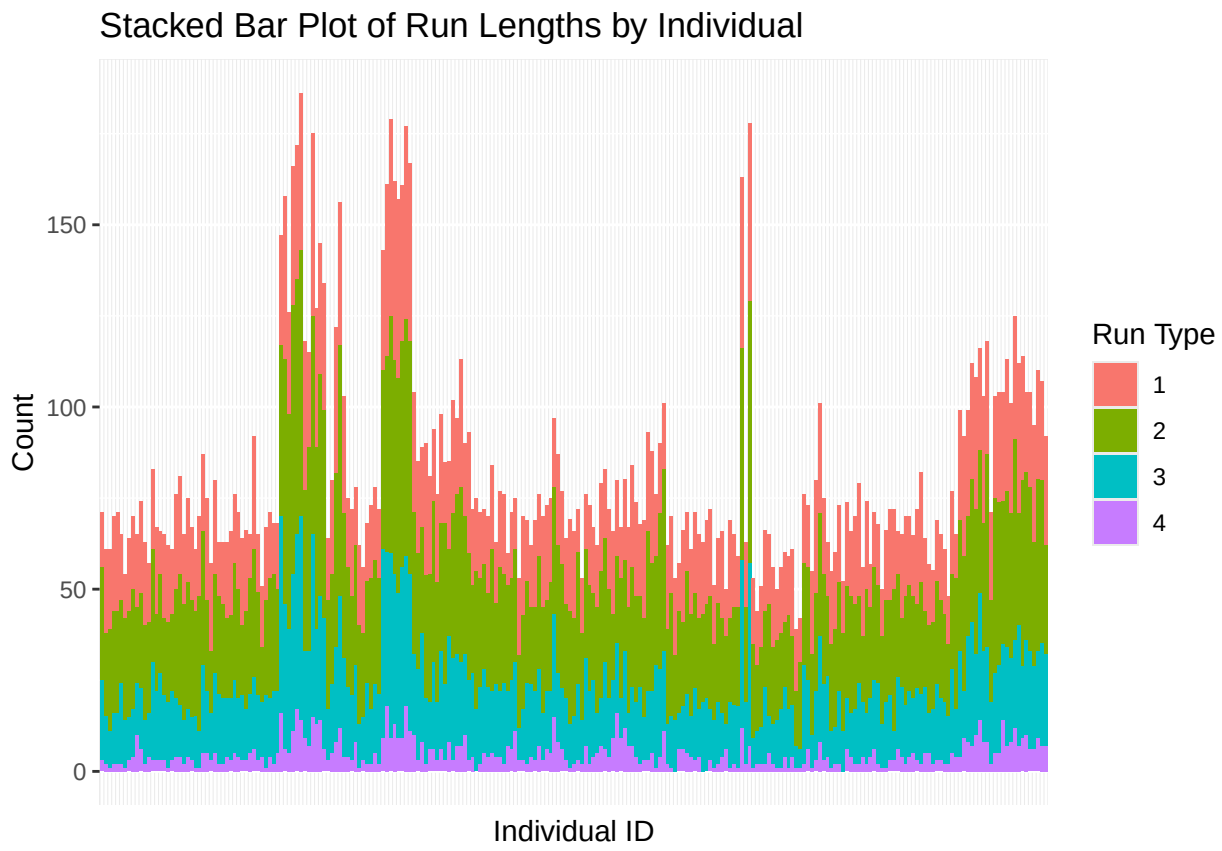



```
runs_per_indiv$population <- ifelse(runs_per_indiv$id %in% merina$id,
  "Merina", runs_per_indiv$population
)

ggplot(runs_per_indiv, aes(x = population, y = n, fill = classification)) +
  geom_bar(stat = "identity", position = "stack") +
  labs(
    title = "Stacked Bar Plot of Run Lengths by Population",
    x = "Individual ID",
    y = "Count",
    fill = "Run Type"
  ) +
  # 90 degree hjust on axis text
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```



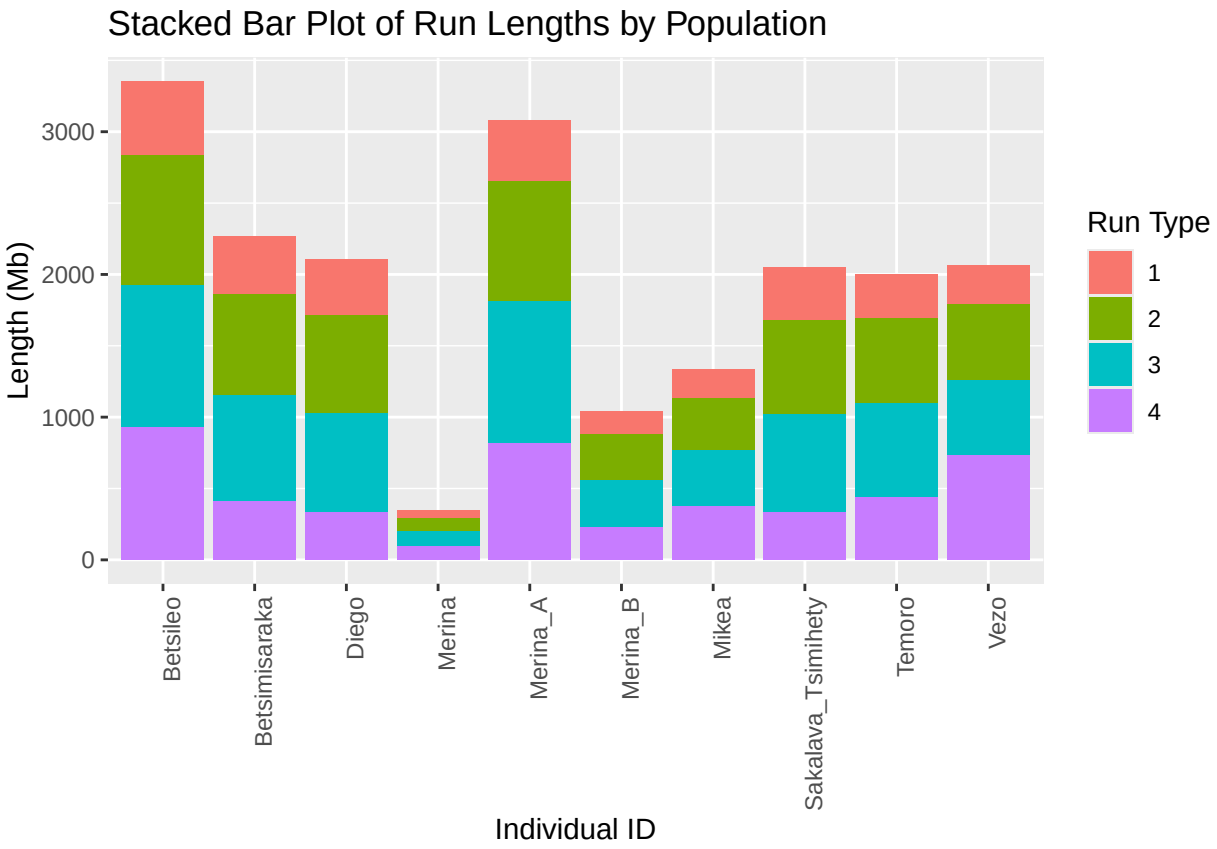

```
ggplot(runs_per_indiv, aes(x = id, y = n, fill = classification)) +
  geom_bar(stat = "identity", position = "stack") +
  labs(
    title = "Stacked Bar Plot of Run Lengths by Individual",
    x = "Individual ID",
    y = "Count",
    fill = "Run Type"
  ) +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank())
```




```
lengths_per_indiv <- runs %>%
  group_by(id, classification) %>%
  summarise(length = sum(lengthBps), .groups = "drop")

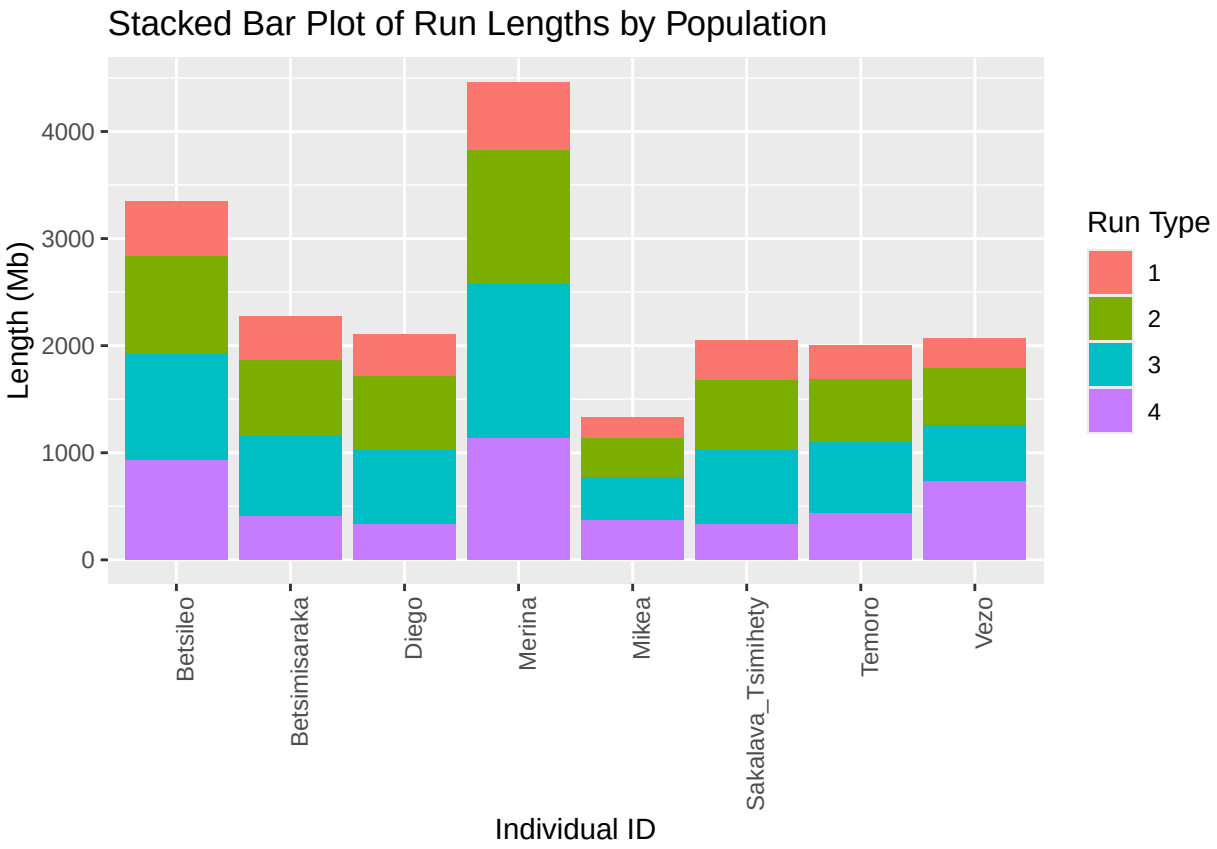
lengths_per_indiv$population <- runs$Population[match(lengths_per_indiv$id, runs$id)]

ggplot(lengths_per_indiv, aes(x = population, y = length / 1000000, fill = classification)) +
  geom_bar(stat = "identity", position = "stack") +
  labs(
    title = "Stacked Bar Plot of Run Lengths by Population",
    x = "Individual ID",
    y = "Length (Mb)",
    fill = "Run Type"
  ) +
  # 90 degree hjust on axis text
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```

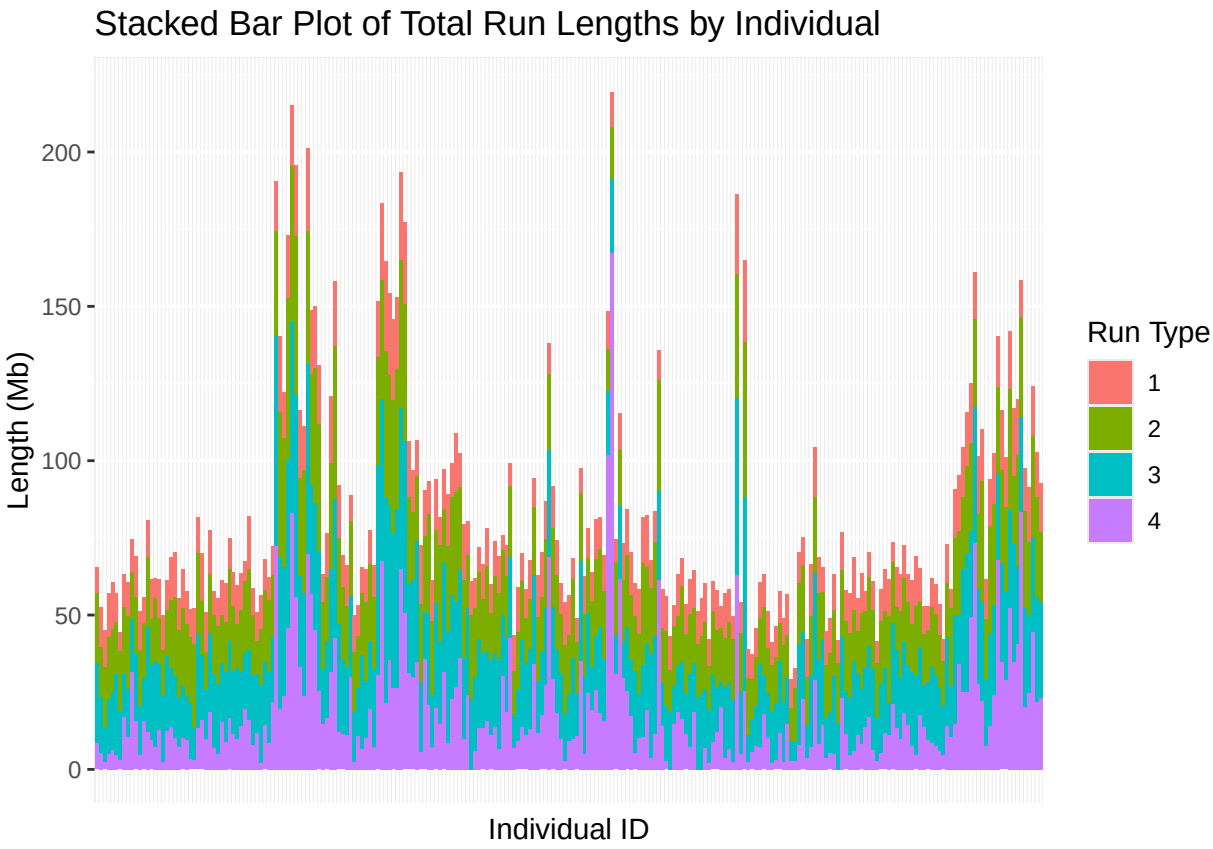



```
lengths_per_indiv$population <- ifelse(lengths_per_indiv$id %in% merina$id,
  "Merina", lengths_per_indiv$population
)

ggplot(lengths_per_indiv, aes(x = population, y = length / 1000000, fill = classification)) +
  geom_bar(stat = "identity", position = "stack") +
  labs(
    title = "Stacked Bar Plot of Run Lengths by Population",
    x = "Individual ID",
    y = "Length (Mb)",
    fill = "Run Type"
  ) +
  # 90 degree hjust on axis text
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```




```
ggplot(lengths_per_indiv, aes(x = id, y = length / 1000000, fill = classification)) +
  geom_bar(stat = "identity", position = "stack") +
  labs(
    title = "Stacked Bar Plot of Total Run Lengths by Individual",
    x = "Individual ID",
    y = "Length (Mb)",
    fill = "Run Type"
  ) +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank())
```

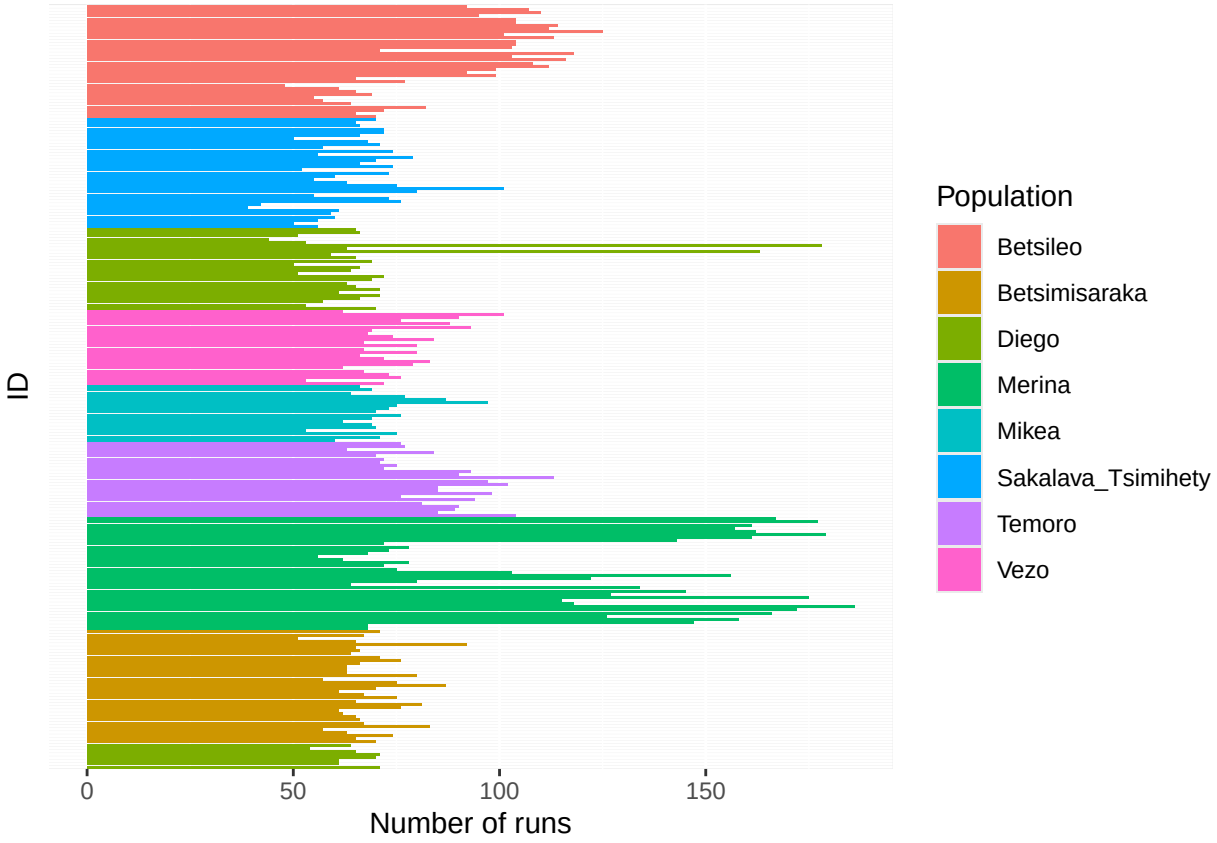


Total number of runs

```
runs_per_indiv <- runs %>%
  group_by(id) %>%
  count()

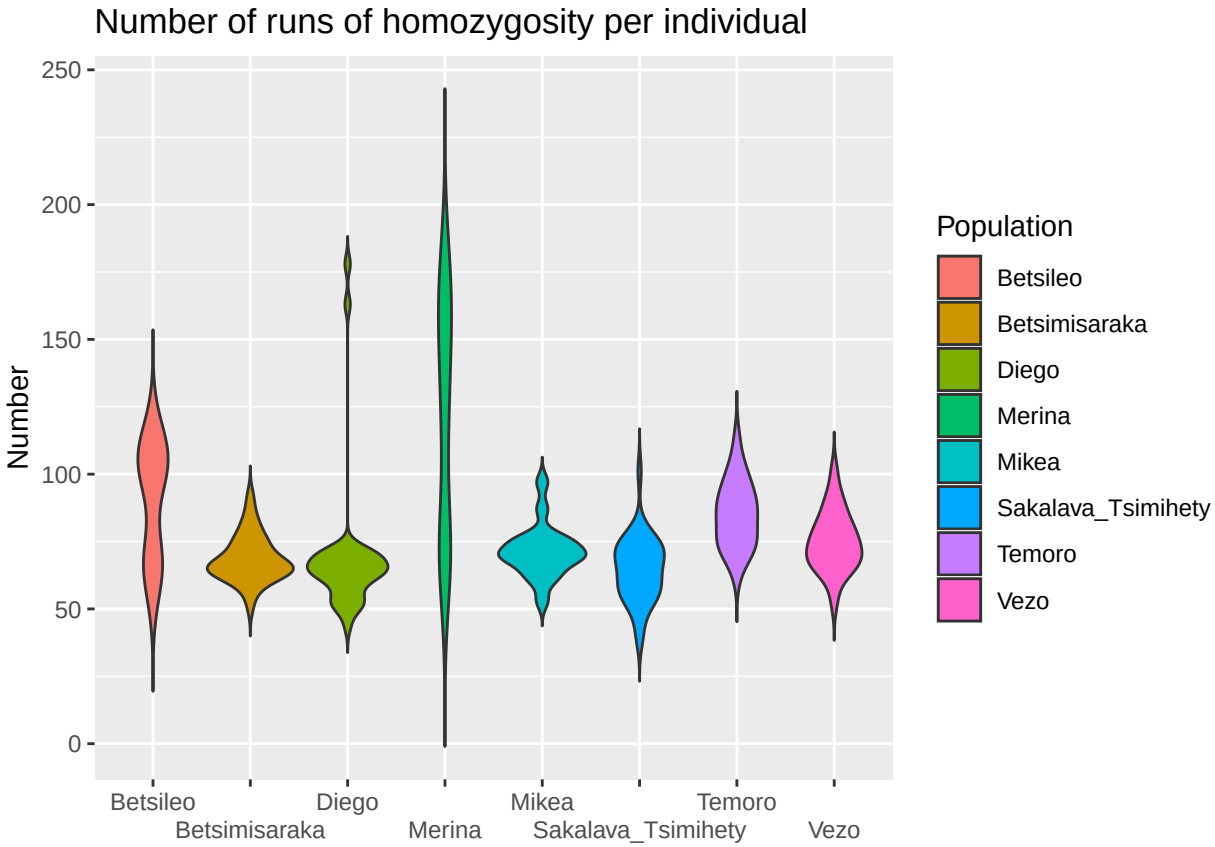
malagasy <- left_join(malagasy, runs_per_indiv, by = "id")
malagasy$n <- as.numeric(malagasy$n)
malagasy$population <- as.factor(malagasy$population)
```

```
ggplot(data = malagasy, aes(x = id, y = n, fill = population)) +
  geom_bar(stat = "identity") +
  coord_flip() +
   theme(
  #   axis.text.x = element_blank(),
  #   axis.ticks.x = element_blank(),
     axis.text.y = element_blank(),
     axis.ticks.y = element_blank()
   ) +
  labs(x = "ID", y = "Number of runs", fill = "Population")
```




```
ggsave(paste0(res_dir, "n_runs_per_indiv.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
ggplot(malagasy, aes(x = population, y = n, fill = population)) +
  geom_violin(trim = FALSE) +
  labs(
    title = "Number of runs of homozygosity per individual",
    y = "Number", x = NULL, fill = "Population"
  ) +
  scale_x_discrete(guide = guide_axis(n.dodge = 2))
```

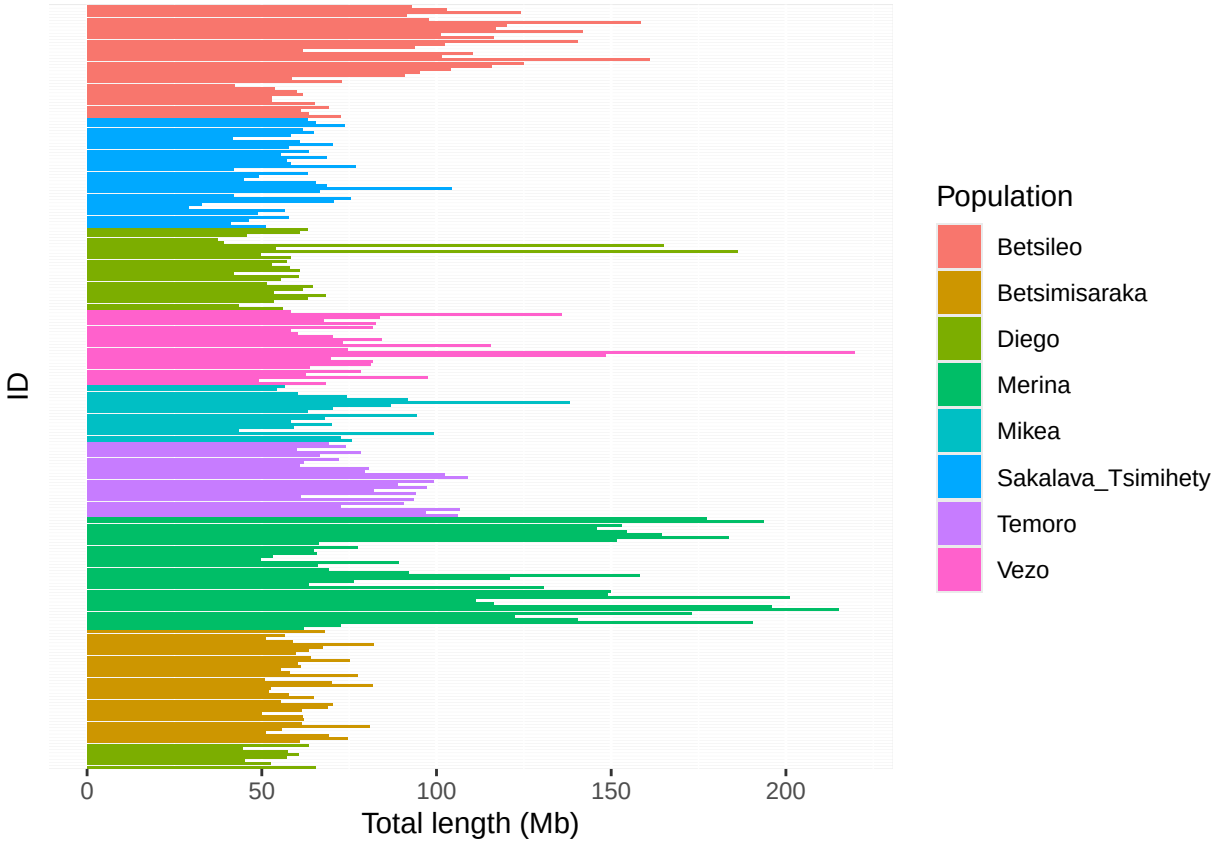



```
ggsave(paste0(res_dir, "n_runs_per_indiv_violin.jpg"),
  width = 7, height = 7, dpi = 300
)
```

Total length of runs

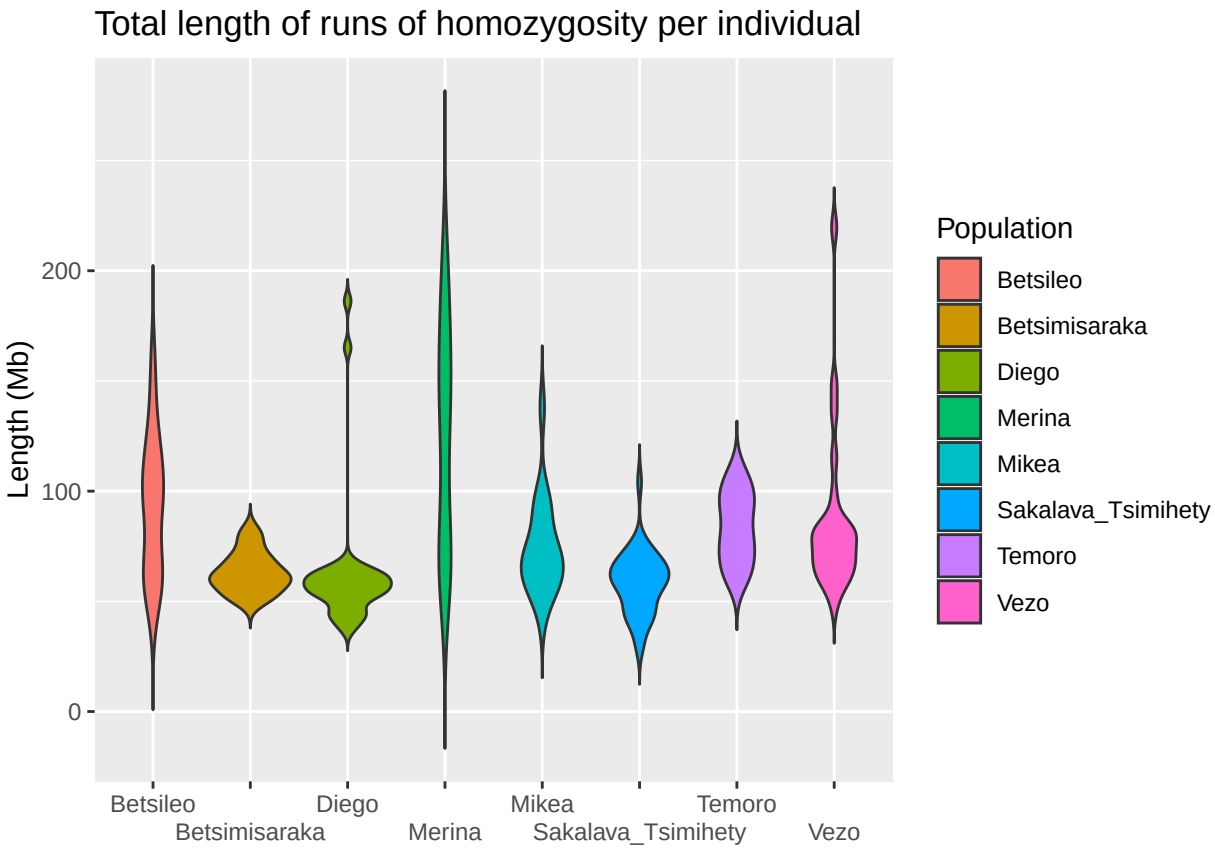
```
total_length_per_indiv <-
  aggregate(runs$lengthBps, by = list(id = runs$id), FUN = sum)
colnames(total_length_per_indiv) <- c("id", "length")
malagasy <- left_join(malagasy, total_length_per_indiv, by = "id")
```

```
ggplot(data = malagasy, aes(x = id, y = length / 1000000, fill = population)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  theme(
    #axis.text.x = element_blank(), axis.ticks.x = element_blank(),
    axis.text.y = element_blank(), axis.ticks.y = element_blank()
  ) +
  labs(x = "ID", y ="Total length (Mb)", fill = "Population")
```




```
ggsave(paste0(res_dir, "total_len_per_indiv.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
ggplot(malagasy, aes(x = population, y = length / 1000000, fill = population)) +
  geom_violin(trim = FALSE) +
  labs(
    title = "Total length of runs of homozygosity per individual",
    y = "Length (Mb)", x = NULL, fill = "Population"
  ) +
  scale_x_discrete(guide = guide_axis(n.dodge = 2))
```

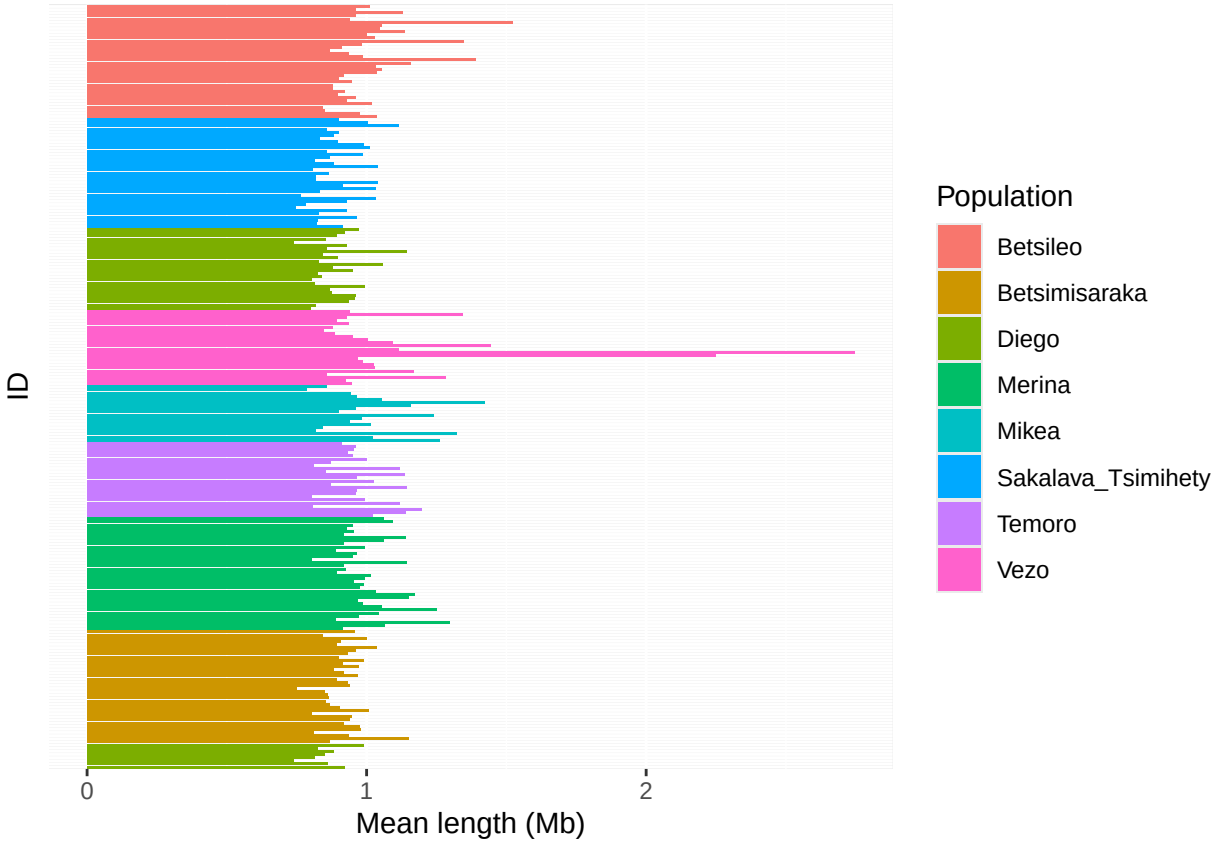



```
ggsave(paste0(res_dir, "total_len_per_indiv_violin.jpg"),
  width = 7, height = 7, dpi = 300
)
```

Average length of runs per individual

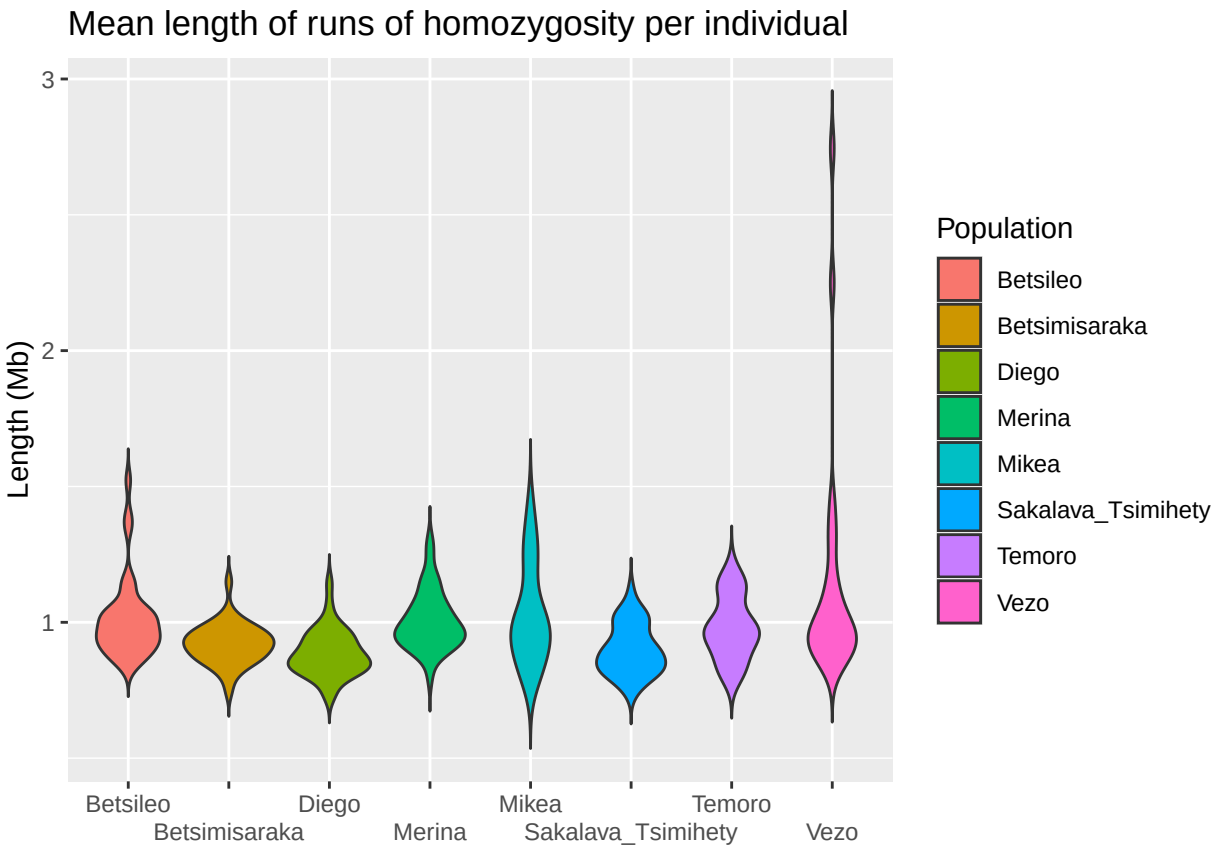
```
avg_length_per_indiv <- aggregate(runs$lengthBps,
  by = list(id = runs$id), FUN = mean
)
colnames(avg_length_per_indiv) <- c("id", "mean")
malagasy <- left_join(malagasy, avg_length_per_indiv, by = "id")
```

```
ggplot(data = malagasy, aes(x = id, y = mean / 1000000, fill = population)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  theme(
  #  axis.text.x = element_blank(), axis.ticks.x = element_blank(),
    axis.text.y = element_blank(), axis.ticks.y = element_blank()
  ) +
  labs(x =  "ID", y ="Mean length (Mb)", fill = "Population")
```




```
ggsave(paste0(res_dir, "mean_len_per_indiv.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
ggplot(malagasy, aes(x = population, y = mean / 1000000, fill = population)) +
  geom_violin(trim = FALSE) +
  labs(
    title = "Mean length of runs of homozygosity per individual",
    y = "Length (Mb)", x = NULL, fill = "Population"
  ) +
  scale_x_discrete(guide = guide_axis(n.dodge = 2))
```

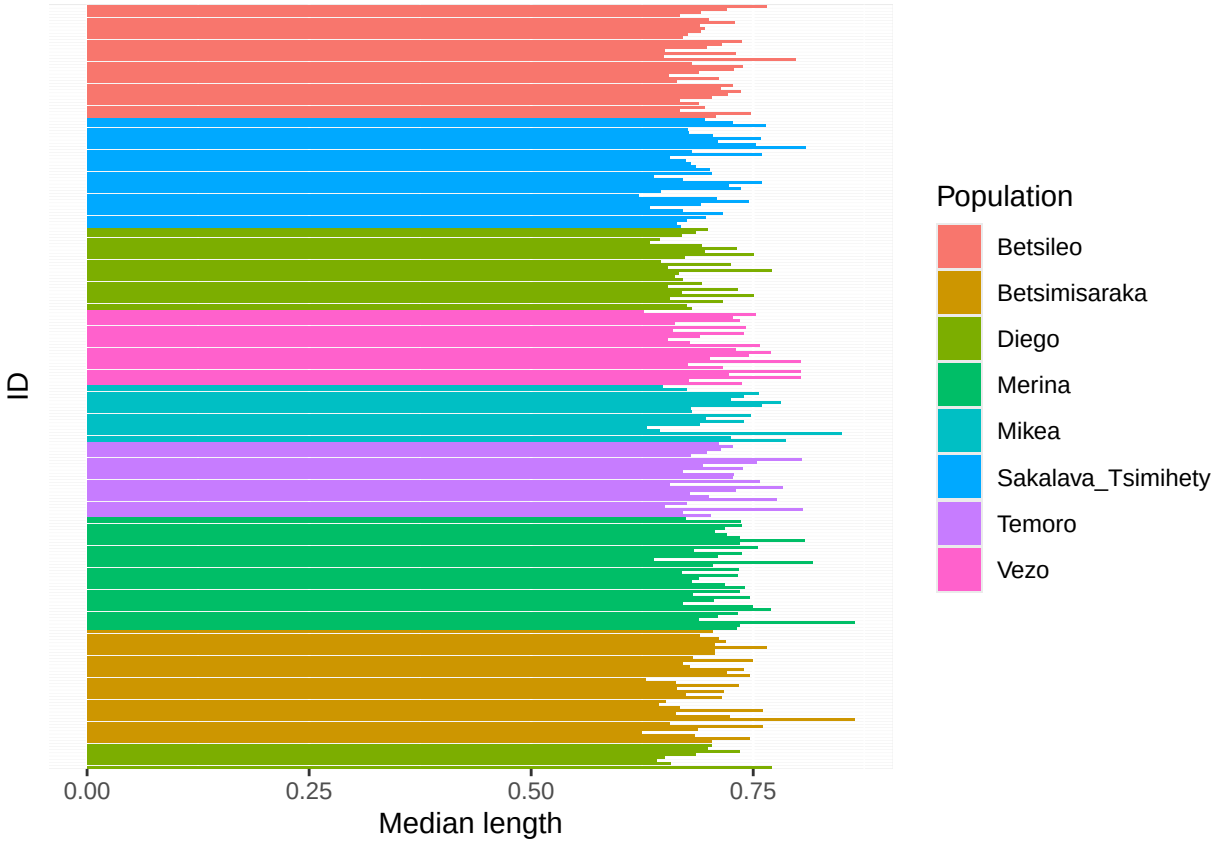



```
ggsave(paste0(res_dir, "mean_len_per_indiv_violin.jpg"),
  width = 7, height = 7, dpi = 300
)
```

Median length of runs per individual

```
avg_length_per_indiv <- aggregate(runs$lengthBps,
  by = list(id = runs$id), FUN = median
)
colnames(avg_length_per_indiv) <- c("id", "median")
malagasy <- left_join(malagasy, avg_length_per_indiv, by = "id")
```

```
ggplot(data = malagasy, aes(x = id, y = median / 1000000, fill = population)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  theme(
   # axis.text.x = element_blank(), axis.ticks.x = element_blank(),
    axis.text.y = element_blank(), axis.ticks.y = element_blank()
  ) +
  labs(x = "ID", y = "Median length", fill = "Population")
```

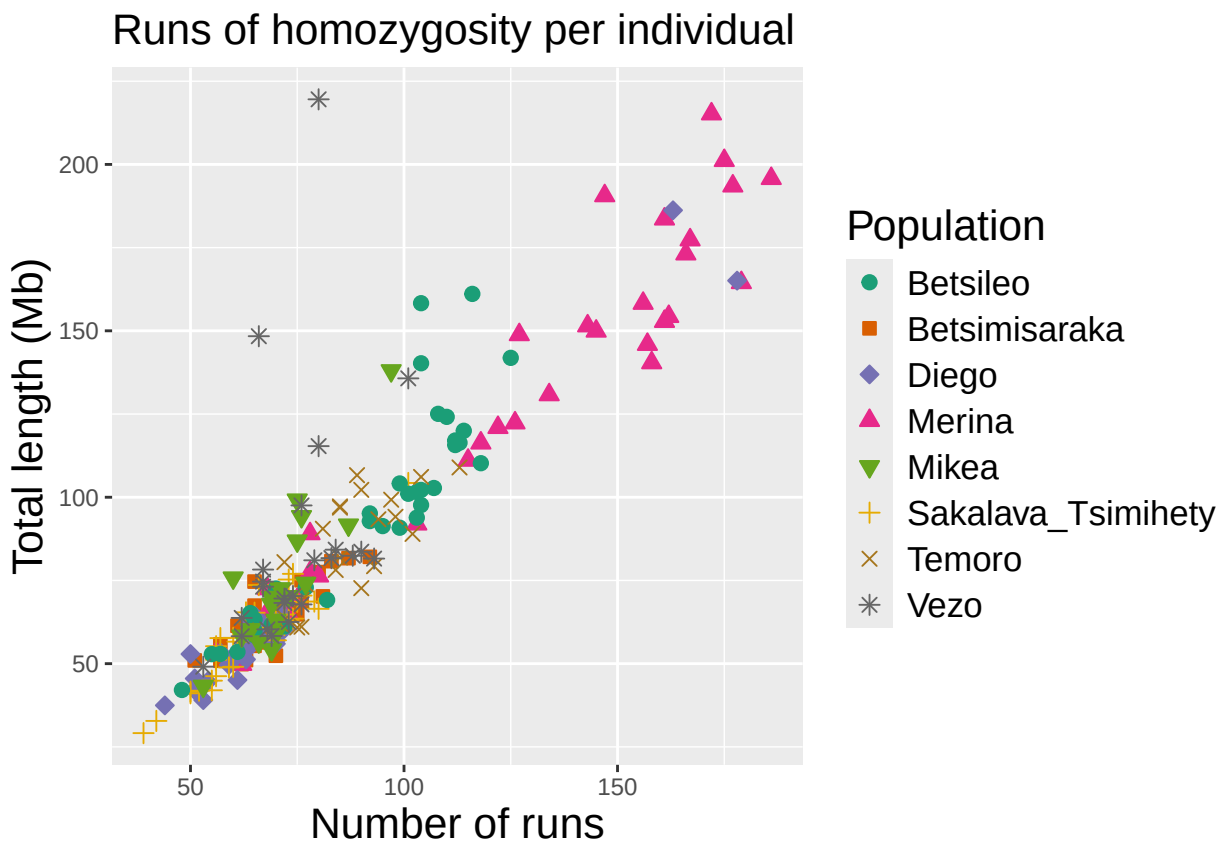



```
ggsave(paste0(res_dir, "mean_len_per_indiv.jpg"),
  width = 7, height = 7, dpi = 300
)
```

Total length by total number

```
ggplot(malagasy, aes(
  x = n, y = length / 1000000, color = population,
  fill = population, shape = population
)) +
  geom_point(size = 2.2) +
  labs(
    title = "Runs of homozygosity per individual",
    x = "Number of runs", y = "Total length (Mb)",
    color = "Population", shape = "Population",
    fill = "Population"
  ) +
  scale_color_brewer(palette = "Dark2") +
  scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
  theme(
    plot.title = element_text(size = 16),
    axis.title.x = element_text(size = 16),
    axis.title.y = element_text(size = 16),
    legend.title = element_text(size = 16),
    legend.text = element_text(size = 13)
```

```
  ) +
  scale_fill_brewer(palette = "Dark2")
```

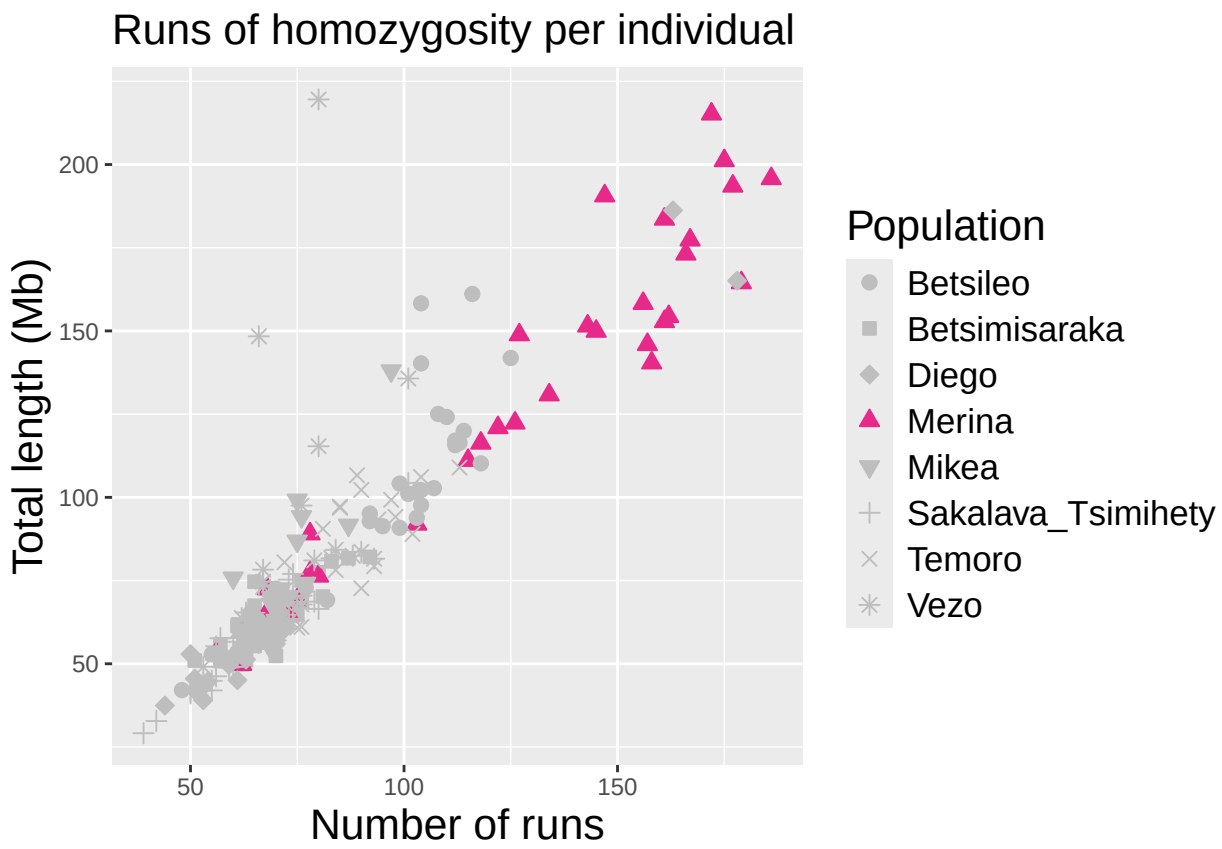



```
ggsave(paste0(res_dir, "total_len_vs_n_runs.jpg"), width = 9, height = 7, dpi = 300)
```

```
# same plot, but grey out all but the Merina
malagasy$merina <- "Other"
malagasy$merina <- ifelse(malagasy$population == "Merina",
  "Merina", malagasy$merina
)

ggplot(malagasy, aes(
  x = n, y = length / 1000000, color = merina,
  fill = merina, shape = population
)) +
  geom_point(size = 2.2) +
  labs(
    title = "Runs of homozygosity per individual",
    x = "Number of runs", y = "Total length (Mb)",
    color = NULL, shape = "Population", fill = NULL
  ) +
  scale_fill_manual(values = c("Merina" = "#E7298A", "Other" = "grey")) +
  scale_color_manual(values = c("Merina" = "#E7298A", "Other" = "grey")) +
  scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
  guides(
    fill = "none", color = "none",
```

```
    shape = guide_legend(override.aes = list(
      fill = c("grey", "grey", "grey", "#E7298A", "grey", "grey", "grey", "grey"),
      color = c("grey", "grey", "grey", "#E7298A", "grey", "grey", "grey", "grey")
    ))
  ) +
  theme(
    plot.title = element_text(size = 16),
    axis.title.x = element_text(size = 16),
    axis.title.y = element_text(size = 16),
    legend.title = element_text(size = 16),
    legend.text = element_text(size = 13)
  )
```

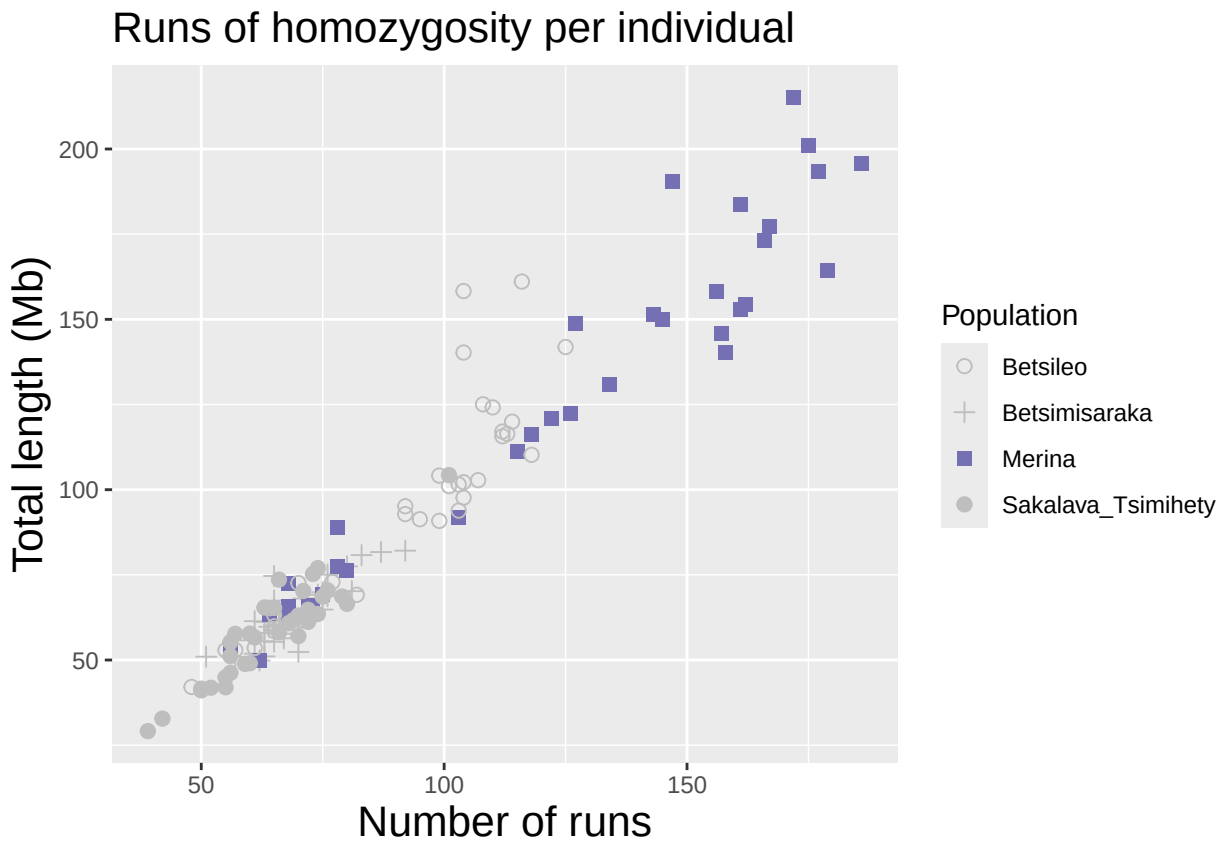



```
ggsave(paste0(res_dir, "total_len_vs_n_runs_merina.jpg"),
  width = 9, height = 7, dpi = 300
)
```

```
#
malagasy_subset <- malagasy[malagasy$population %in% c("Betsileo", "Merina", "Betsimisaraka", "Sakalava_

ggplot(malagasy_subset, aes(x = n, y = length / 1000000, color = merina, fill = merina, shape = populati
  geom_point(size = 2.2) +
  labs(title = "Runs of homozygosity per individual", x = "Number of runs", y = "Total length (Mb)", col
  scale_fill_manual(values = c("Merina" = "#7570B3", "Other" = "grey")) +
  scale_color_manual(values = c("Merina" = "#7570B3", "Other" = "grey")) +
  # scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
```

```
  scale_shape_manual(values = c(1, 3, 15, 19)) +
  guides(
    fill = "none", color = "none",
    shape = guide_legend(override.aes = list(
      fill = c("grey", "grey", "#7570B3", "grey"),
      color = c("grey", "grey", "#7570B3", "grey")
    ))
  ) +
  theme(
    plot.title = element_text(size = 16),
    axis.title.x = element_text(size = 16),
    axis.title.y = element_text(size = 16)
  )
```




```
ggsave("/home/EJC199/Dropbox/malagasy_assortative_mating/ROH.jpg",
  width = 7, height = 5, dpi = 500
)
```

Clear differences are observed between individuals and ethnic groups in the number and length of runs of homozygosity. Several factors explain this pattern:

1) The African component of Malagasy ancestry will be likely to have fewer ROH's, as African populations have lower homozygosity than other global pops, which introduces variation between individuals based on their proportion of African ancestry.

2) IBD analysis (Alva et al 2022) suggests a bottleneck in the Asian ancestry source of all Malagasy people, meaning (if all other things were equal) the Asian ancestry component would have more ROH's than

the African component

Kruskal Wallis test

```
kruskal.test(length ~ population, data = malagasy)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  length by population
## Kruskal-Wallis chi-squared = 92.4, df = 7, p-value < 2.2e-16
```

Wilcox test

```
pairwise.wilcox.test(malagasy$length, malagasy$population, p.adjust.method = "bonferroni")
```

```
##
##  Pairwise comparisons using Wilcoxon rank sum exact test
##
## data:  malagasy$length and malagasy$population
##
##                    Betsileo Betsimisaraka Diego   Merina  Mikea
## Betsimisaraka      0.00033  -             -       -       -
## Diego              2.1e-05  0.27953       -       -       -
## Merina             0.18115  1.6e-07       5.8e-09 -       -
## Mikea              0.92815  1.00000       0.02587 0.01549 -
## Sakalava_Tsimihety 4.2e-05  1.00000       1.00000 1.0e-08 0.20833
## Temoro             1.00000  2.7e-05       5.0e-07 0.17085 1.00000
## Vezo               1.00000  0.00445       2.2e-05 0.20702 1.00000
##                    Sakalava_Tsimihety Temoro
## Betsimisaraka      -                  -
## Diego              -                  -
## Merina             -                  -
## Mikea              -                  -
## Sakalava_Tsimihety -                  -
## Temoro             3.7e-06            -
## Vezo               0.00027            1.00000
##
## P value adjustment method: bonferroni
```

To see how much ancestry is related to runs of homozygosity, we can plot correlations of ROH versus ancestry for all individuals.

Load q_matrices from fastmixture

```
qmats <- readRDS("../data/intermediate/fastmixture/fastmixture_m.RData")
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.gt")
```

```
tidyq <- augment(get_q_matrix(qmats, k = 3, run = 1),
  data = mad_bor_g1000_sum
```

```
)
head(tidyq)
```

```
## # A gen_tibble: 498926 loci
## # A tibble:     6 x 7
##   id       population     genotypes .rownames .Q1        .Q2        .Q3
##   <chr>    <chr>         <vctr_SNP> <chr>     <q_matrix> <q_matrix> <q_matrix>
## 1 BET001RR Betsimisaraka          1 BET001RR  0.3345989  0.6363412  0.02905989
## 2 BET002RR Betsimisaraka          2 BET002RR  0.3230504  0.6401900  0.03675959
## 3 BET003RR Betsimisaraka          3 BET003RR  0.3173466  0.6623318  0.02032159
## 4 BET004RR Betsimisaraka          4 BET004RR  0.3581365  0.5945273  0.04733628
## 5 BET006RR Betsimisaraka          5 BET006RR  0.3056575  0.6559178  0.03842471
## 6 BET007RR Betsimisaraka          6 BET007RR  0.3354124  0.6386873  0.02590029
```

```
all_malagasy <- c(
  "Betsimisaraka", "Merina", "Sakalava_Tsimihety",
  "Betsileo", "Diego", "Vezo", "Mikea", "Temoro"
)

malagasyq <- tidyq %>% filter(population %in% all_malagasy)
head(malagasyq)
```

```
## # A gen_tibble: 498926 loci
## # A tibble:     6 x 7
##   id       population     genotypes .rownames .Q1        .Q2        .Q3
##   <chr>    <chr>         <vctr_SNP> <chr>     <q_matrix> <q_matrix> <q_matrix>
## 1 BET001RR Betsimisaraka          1 BET001RR  0.3345989  0.6363412  0.02905989
## 2 BET002RR Betsimisaraka          2 BET002RR  0.3230504  0.6401900  0.03675959
## 3 BET003RR Betsimisaraka          3 BET003RR  0.3173466  0.6623318  0.02032159
## 4 BET004RR Betsimisaraka          4 BET004RR  0.3581365  0.5945273  0.04733628
## 5 BET006RR Betsimisaraka          5 BET006RR  0.3056575  0.6559178  0.03842471
## 6 BET007RR Betsimisaraka          6 BET007RR  0.3354124  0.6386873  0.02590029
```

Comparing proportion to ROH's

```
malagasy_all <- malagasyq

# match
matching_m <- match(malagasyq$id, malagasy$id)
malagasy_all$n <- "NA"
malagasy_all$length <- "NA"
malagasy_all$mean <- "NA"
malagasy_all$n <- ifelse(!is.na(matching_m), malagasy$n[matching_m], malagasy_all$n)
malagasy_all$length <- ifelse(!is.na(matching_m), malagasy$length[matching_m], malagasy_all$length)
malagasy_all$mean <- ifelse(!is.na(matching_m), malagasy$mean[matching_m], malagasy_all$mean)
malagasy_all$median <- ifelse(!is.na(matching_m), malagasy$median[matching_m], malagasy_all$median)
head(malagasy_all)
```

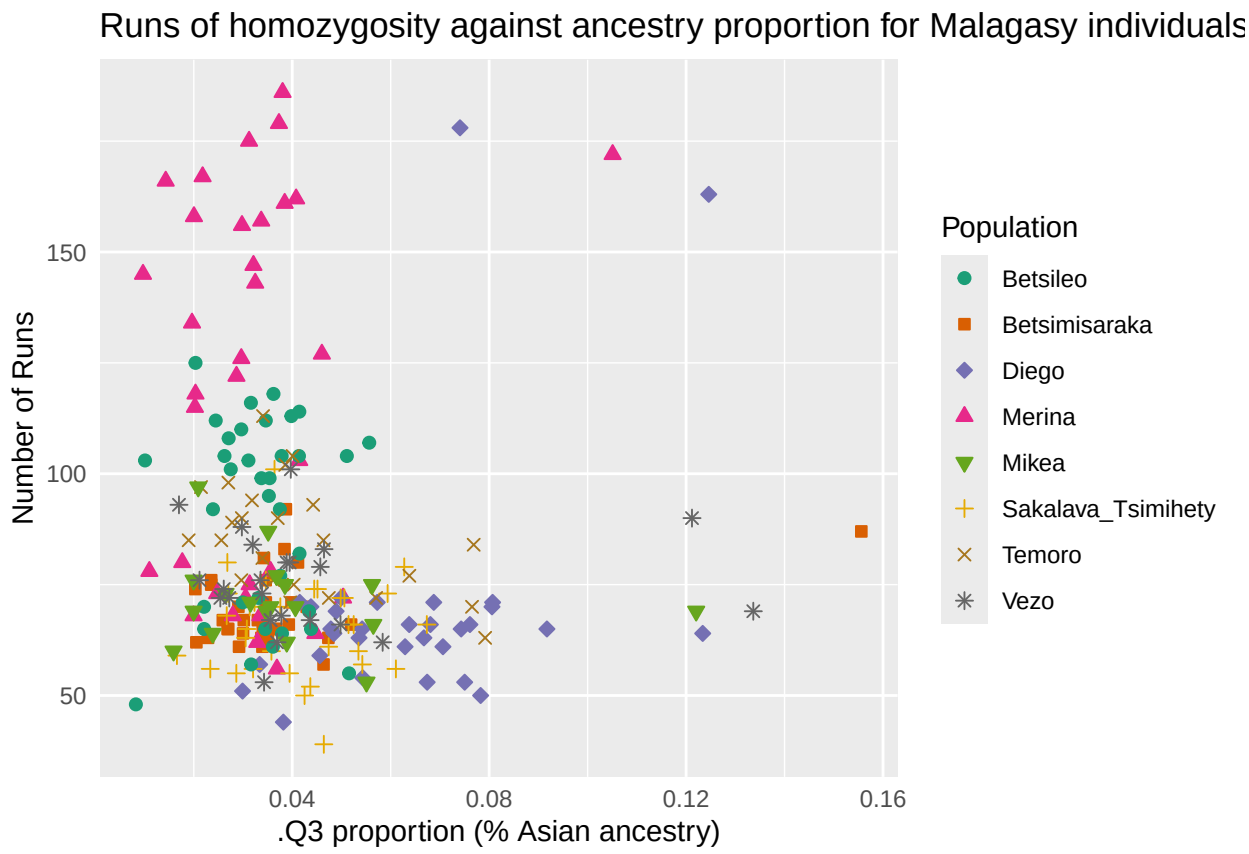
```
## # A gen_tibble: 498926 loci
## # A tibble:     6 x 11
##   id       population  genotypes .rownames .Q1    .Q2    .Q3       n length   mean
##   <chr>    <chr>       <vctr_SN> <chr>     <q_m> <q_m> <q_m> <dbl>  <int>  <dbl>
## 1 BET001RR Betsimisar~         1 BET001RR  0.33~ 0.63~ 0.02~    70 6.07e7 8.67e5
## 2 BET002RR Betsimisar~         2 BET002RR  0.32~ 0.64~ 0.03~    65 7.47e7 1.15e6
```

```
## 3 BET003RR Betsimisar~         3 BET003RR  0.31~ 0.66~ 0.02~    74 6.92e7 9.35e5
## 4 BET004RR Betsimisar~         4 BET004RR  0.35~ 0.59~ 0.04~    63 5.11e7 8.11e5
## 5 BET006RR Betsimisar~         5 BET006RR  0.30~ 0.65~ 0.03~    83 8.08e7 9.73e5
## 6 BET007RR Betsimisar~         6 BET007RR  0.33~ 0.63~ 0.02~    67 6.15e7 9.18e5
## # i 1 more variable: median <dbl>
```

Ancestry proportion compared to number of ROHS

```
ggplot(malagasy_all, aes(
  x = .Q3, y = n,
  fill = population,
  colour = population, shape = population
)) +
  geom_point(size = 1.8) +
  labs(
    title = "Runs of homozygosity against ancestry proportion for Malagasy individuals",
    x = ".Q3 proportion (% Asian ancestry)", y = "Number of Runs"
  ) +
  scale_color_brewer(palette = "Dark2") +
  scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
  theme(axis.ticks.x = element_blank(), axis.ticks.y = element_blank()) +
  # axis.text.x = element_blank(), axis.text.y = element_blank(),
  scale_fill_brewer(palette = "Dark2") +
  guides(
    fill = guide_legend(title = "Population"),
    colour = guide_legend(title = "Population"),
    shape = guide_legend(title = "Population")
  )
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```

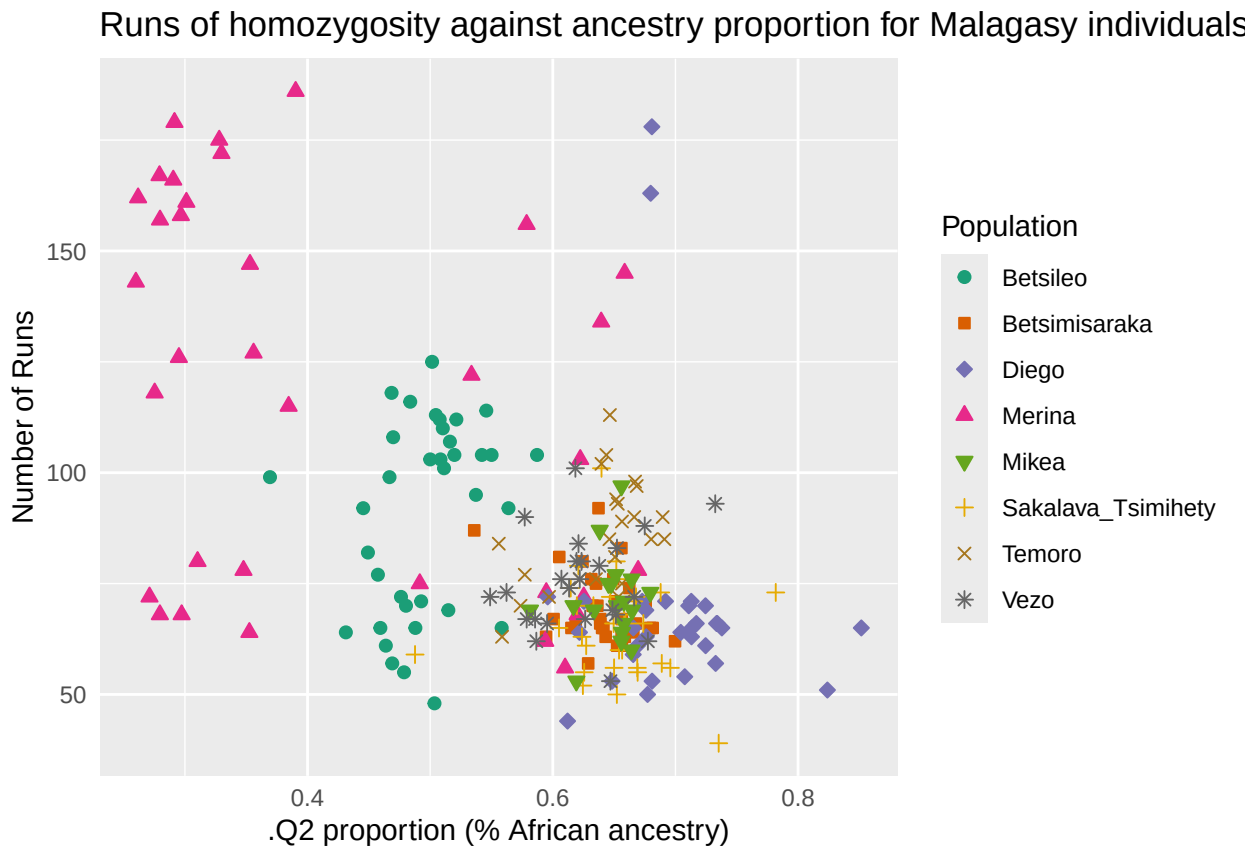



```
ggsave(paste0(res_dir, "n_runs_vs_asian_ancestry.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```

```
ggplot(malagasy_all, aes(
  x = .Q2, y = n, fill = population,
  colour = population, shape = population
)) +
  geom_point(size = 1.8) +
  labs(
    title = "Runs of homozygosity against ancestry proportion for Malagasy individuals",
    x = ".Q2 proportion (% African ancestry)", y = "Number of Runs"
  ) +
  scale_color_brewer(palette = "Dark2") +
  scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
  theme(axis.ticks.x = element_blank(), axis.ticks.y = element_blank()) +
  # axis.text.x = element_blank(),axis.text.y = element_blank(),
  scale_fill_brewer(palette = "Dark2") +
  guides(
    fill = guide_legend(title = "Population"),
    colour = guide_legend(title = "Population"),
    shape = guide_legend(title = "Population")
  )
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```




```
ggsave(paste0(res_dir, "n_runs_vs_african_ancestry.jpg"),
  width = 7, height = 7, dpi = 300
)
```

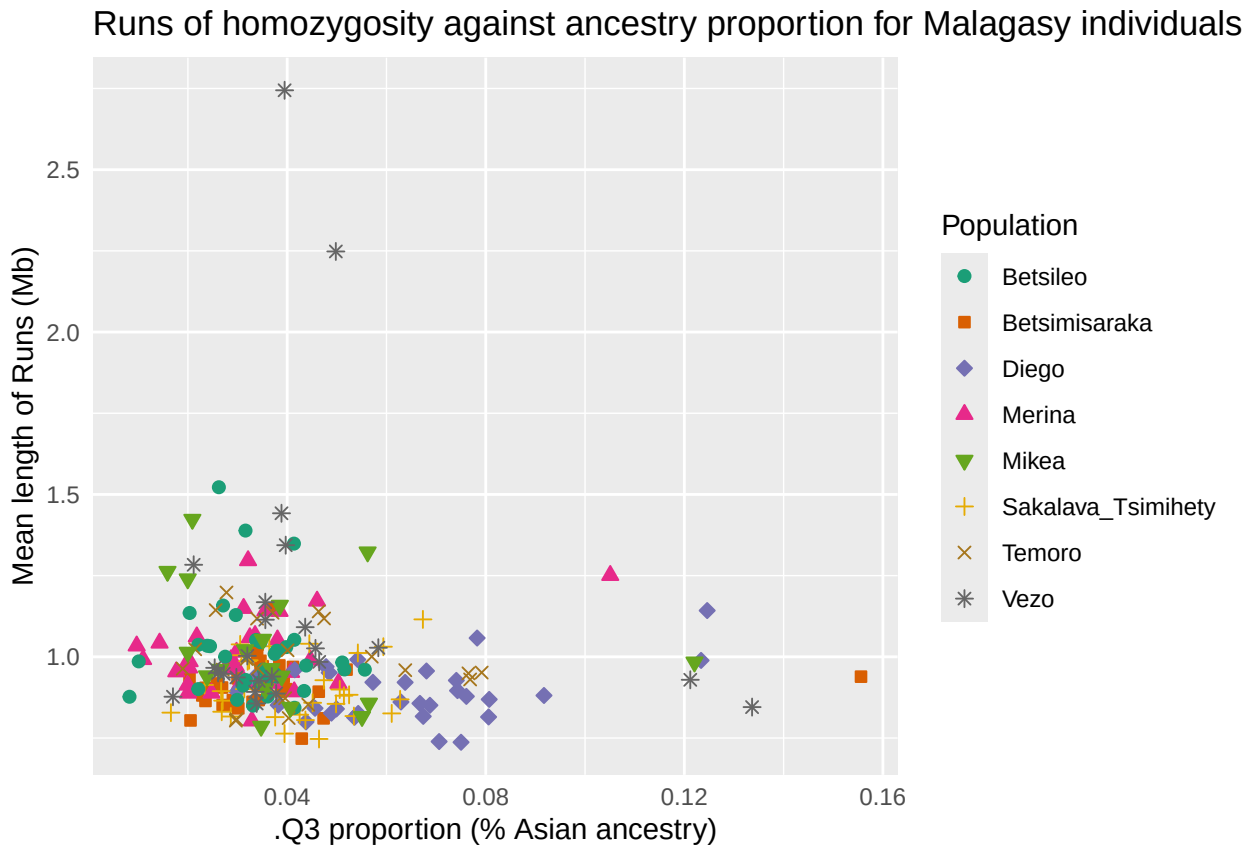
```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```

Ancestry proportion compared to mean length of ROHS

```
ggplot(malagasy_all, aes(
  x = .Q3, y = mean / 1000000, fill = population,
  colour = population, shape = population
)) +
  geom_point(size = 1.8) +
  labs(
    title = "Runs of homozygosity against ancestry proportion for Malagasy individuals",
    x = ".Q3 proportion (% Asian ancestry)", y = "Mean length of Runs (Mb)"
  ) +
  scale_color_brewer(palette = "Dark2") +
  scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
  theme(axis.ticks.x = element_blank(), axis.ticks.y = element_blank()) +
  # axis.text.x = element_blank(),  axis.text.y = element_blank(),
  scale_fill_brewer(palette = "Dark2") +
```

```
  guides(
    fill = guide_legend(title = "Population"),
    colour = guide_legend(title = "Population"),
    shape = guide_legend(title = "Population")
  )
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```




```
ggsave(paste0(res_dir, "mean_len_vs_asian_ancestry.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```

Two Vezo individuals are clear outliers in the above plot. If we remove these individuals what happens?

```
nrow(malagasy_all)
```

```
## [1] 232
```

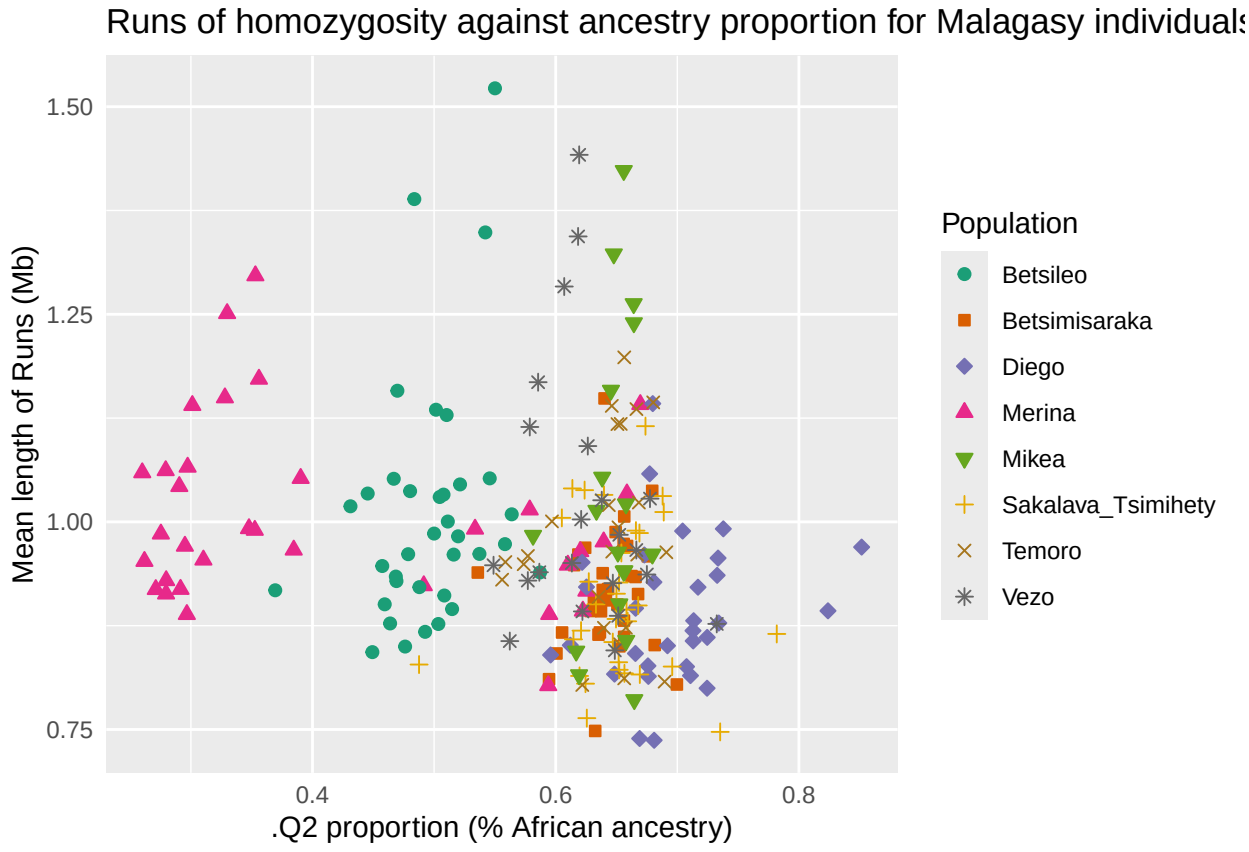
```
malagasy_all <- malagasy_all %>% filter(mean < 2000000)
nrow(malagasy_all)
```

```
## [1] 230
```

```
ggplot(malagasy_all, aes(
  x = .Q2, y = mean / 1000000, fill = population,
```

```
  colour = population, shape = population
)) +
  geom_point(size = 1.8) +
  labs(
    title = "Runs of homozygosity against ancestry proportion for Malagasy individuals",
    x = ".Q2 proportion (% African ancestry)", y = "Mean length of Runs (Mb)"
  ) +
  scale_color_brewer(palette = "Dark2") +
  scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
  theme(axis.ticks.x = element_blank(), axis.ticks.y = element_blank()) +
  # axis.text.x = element_blank(), axis.text.y = element_blank(),
  scale_fill_brewer(palette = "Dark2") +
  guides(
    fill = guide_legend(title = "Population"),
    colour = guide_legend(title = "Population"),
    shape = guide_legend(title = "Population")
  )
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```



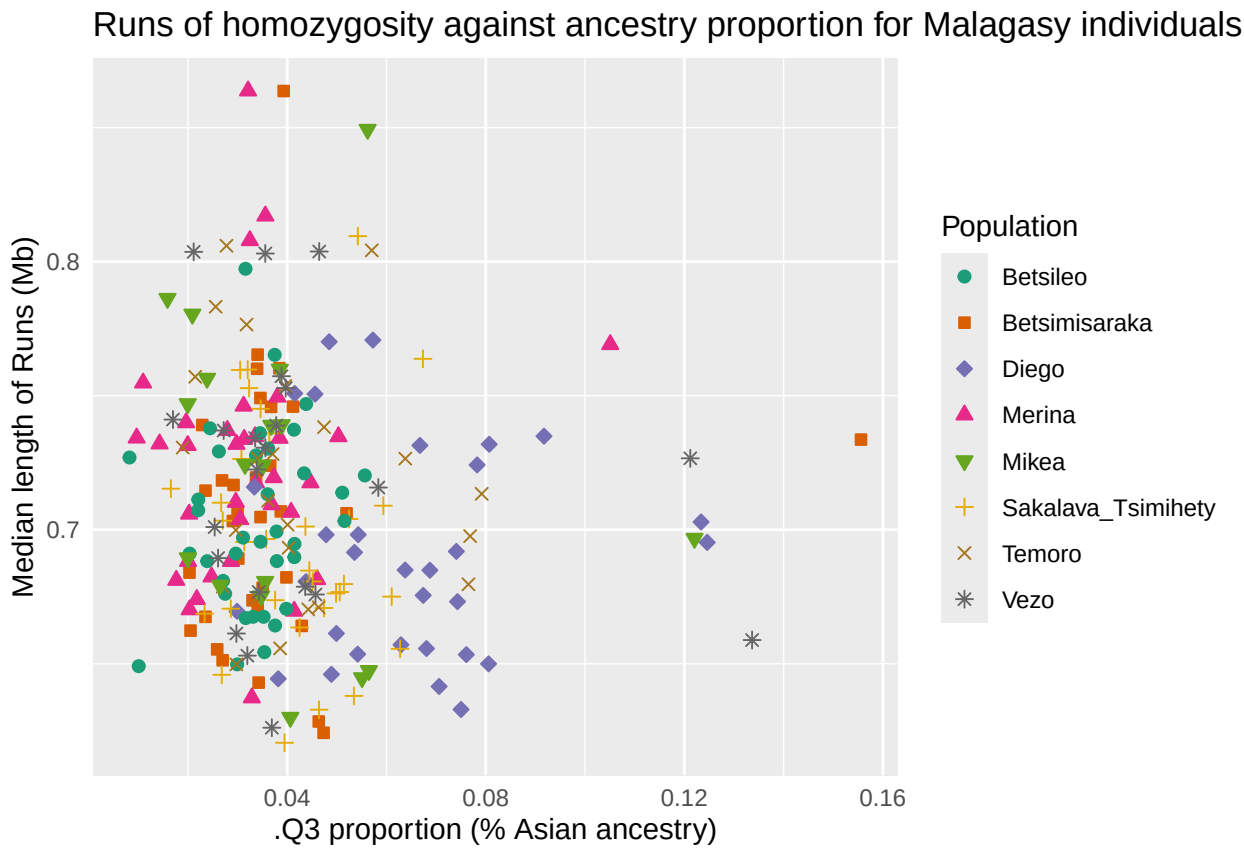

```
ggsave(paste0(res_dir, "mean_len_vs_african_ancestry.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```

Ancestry proportion compared to median length of ROHS

```
ggplot(malagasy_all, aes(
  x = .Q3, y = median / 1000000, fill = population,
  colour = population, shape = population
)) +
  geom_point(size = 1.8) +
  labs(
    title = "Runs of homozygosity against ancestry proportion for Malagasy individuals",
    x = ".Q3 proportion (% Asian ancestry)", y = "Median length of Runs (Mb)"
  ) +
  scale_color_brewer(palette = "Dark2") +
  scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
  theme(axis.ticks.x = element_blank(), axis.ticks.y = element_blank()) +
  # axis.text.x = element_blank(),  axis.text.y = element_blank(),
  scale_fill_brewer(palette = "Dark2") +
  guides(
    fill = guide_legend(title = "Population"),
    colour = guide_legend(title = "Population"),
    shape = guide_legend(title = "Population")
  )
```

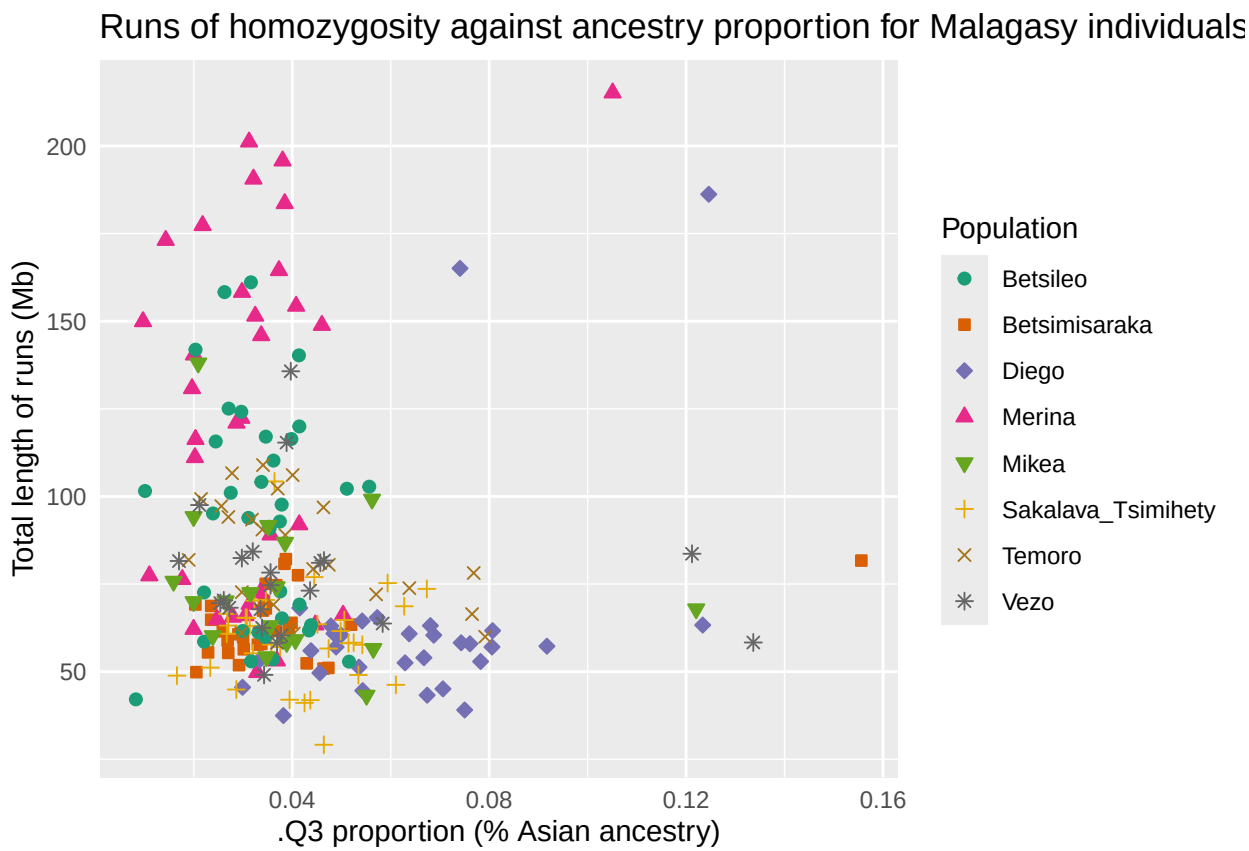
```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```



Ancestry proportion compared to total length of ROHS

```
ggplot(malagasy_all, aes(
  x = .Q3, y = length / 1000000, fill = population,
  colour = population, shape = population
)) +
  geom_point(size = 1.8) +
  labs(
    title = "Runs of homozygosity against ancestry proportion for Malagasy individuals",
    x = ".Q3 proportion (% Asian ancestry)", y = "Total length of runs (Mb)"
  ) +
  scale_color_brewer(palette = "Dark2") +
  scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
  theme(axis.ticks.x = element_blank(), axis.ticks.y = element_blank()) +
  # axis.text.x = element_blank(),  axis.text.y = element_blank(),
  scale_fill_brewer(palette = "Dark2") +
  guides(
    fill = guide_legend(title = "Population"),
    colour = guide_legend(title = "Population"),
    shape = guide_legend(title = "Population")
  )
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```



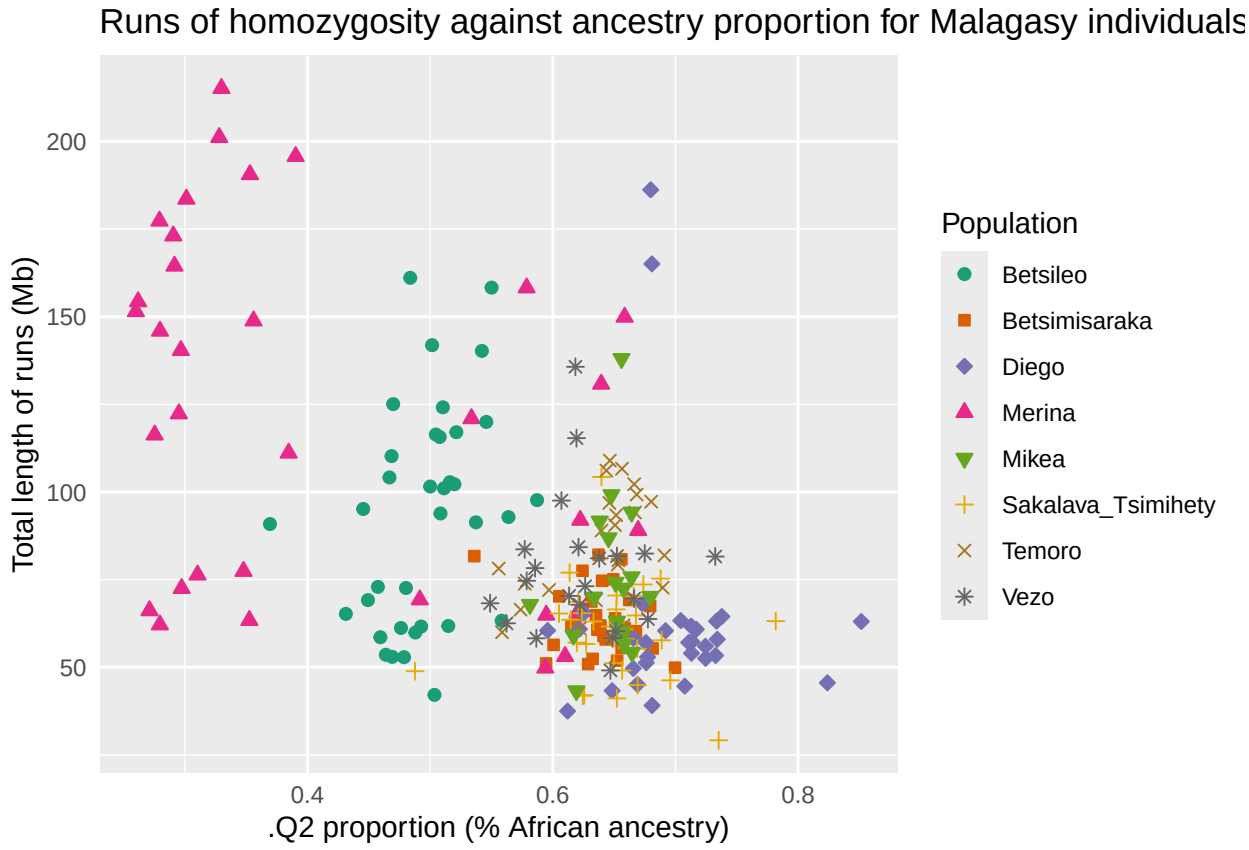

```
ggsave(paste0(res_dir, "total_len_vs_asian_ancestry.jpg"),
```

```
  width = 7, height = 7, dpi = 300
)
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```

```
ggplot(malagasy_all, aes(
  x = .Q2, y = length / 1000000, fill = population,
  colour = population, shape = population
)) +
  geom_point(size = 1.8) +
  labs(
    title = "Runs of homozygosity against ancestry proportion for Malagasy individuals",
    x = ".Q2 proportion (% African ancestry)", y = "Total length of runs (Mb)"
  ) +
  scale_color_brewer(palette = "Dark2") +
  scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
  theme(axis.ticks.x = element_blank(), axis.ticks.y = element_blank()) +
  # axis.text.x = element_blank(),  axis.text.y = element_blank(),
  scale_fill_brewer(palette = "Dark2") +
  guides(
    fill = guide_legend(title = "Population"),
    colour = guide_legend(title = "Population"),
    shape = guide_legend(title = "Population")
  )
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```




```
ggsave(paste0(res_dir, "total_len_vs_african_ancestry.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```

Could we remove the Merina, predict based on the correlation of ancestry/ROH's in the other data where they should fall in ROH distribution given their ancestry, and then show that that model alone doesn't account for the inflation in ROH's?

Kirin, Mirna, Ruth McQuillan, Christopher S. Franklin, Harry Campbell, Paul M. McKeigue, and James F. Wilson. 2010. "Genomic Runs of Homozygosity Record Population History and Consanguinity." *PLOS ONE* 5 (11): e13996. https://doi.org/10.1371/journal.pone.0013996.

Korunes, Katharine L, Giordano Bruno Soares-Souza, Katherine Bobrek, Hua Tang, Isabel Inês Araújo, Amy Goldberg, and Sandra Beleza. 2022. "Sex-Biased Admixture and Assortative Mating Shape Genetic Variation and Influence Demographic Inference in Admixed Cabo Verdeans." *G3 Genes|Genomes|Genetics* 12 (10): jkac183. https://doi.org/10.1093/g3journal/jkac183.

Pemberton, Trevor J., Devin Absher, Marcus W. Feldman, Richard M. Myers, Noah A. Rosenberg, and Jun Z. Li. 2012. "Genomic Patterns of Homozygosity in Worldwide Human Populations." *American Journal of Human Genetics* 91 (2): 275–92. https://doi.org/10.1016/j.ajhg.2012.06.014.

Scrucca, Luca, Chris Fraley, T. Brendan Murphy, and Adrian E. Raftery. 2023. *Model-Based Clustering, Classification, and Density Estimation Using Mclust in r*. New York: Chapman; Hall/CRC. https://doi.org/10.1201/9781003277965.

Szpiech, Zachary A, Alexandra Blant, and Trevor J Pemberton. 2017. "GARLIC: Genomic Autozygosity Regions Likelihood-Based Inference and Classification." *Bioinformatics* 33 (13): 2059–62. https://doi.org/10.1093/bioinformatics/btx102.